

Turbo BEM OTFS Receiver With Optimized Superimposed Pilot Power

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Abstract—In this paper, a Turbo basis expansion modeling (BEM) orthogonal time frequency space (OTFS) scheme is proposed for high-mobility communications subject to either Doppler-shift or Doppler-spread channel. It consists of superimposed pilot power and BEM order optimization and Turbo BEM OTFS receiver. The signal-to-interference-and-noise ratio (SINR) is derived as functions of OTFS system parameters and several channel information (*i.e.*, channel correlation matrix, noise variance, etc.). The optimal superimposed pilot power ratio is then derived with a closed-form solution by calculating the first derivative of derived SINR. BEM order is specially optimized for each channel, instead of simply increasing or reducing it. The Turbo BEM OTFS receiver is proposed with an exchange of soft information between BEM channel estimation, signal detection, and data decoding, leading to high reliability. By using superimposed pilots for initial BEM channel estimation, the proposed Turbo BEM OTFS scheme has zero dedicated pilot overhead, resulting in high spectral efficiency. Simulation results confirm that the proposed OTFS scheme outperforms the existing OTFS schemes in terms of bit error rate (BER) and spectral efficiency. Extrinsic information exchange transfer (EXIT) chart analysis and simulation results also exhibit the fast convergence speed of proposed OTFS scheme.

Index Terms—Orthogonal time frequency space (OTFS), turbo receiver, basis expansion modeling (BEM), superimposed pilot, Doppler spread, Doppler spectrum.

I. INTRODUCTION

THE six generation (6G) wireless communication network is expected to commercialize at around 2030. It is anticipated to achieve higher throughput, higher data rate, enhanced reliability, and lower latency over the fifth

generation (5G) communication network. Orthogonal time frequency space (OTFS) has been regarded as one of important technologies of 6G, thanks to its remarkable performance in high-mobility communications [1], [2], [3], [4], [5], [6]. The research in OTFS has been active in the last few years and several important research topics have been identified, including channel estimation and signal detection [1], [2], transceiver and waveform design [3], OTFS based multiple access schemes [4], [7], OTFS with (massive) antennas [5], [6], [7], machine-learning enhanced OTFS [8], [9], integrated communication and sensing via OTFS [10], [11], etc. The research focus of this paper is on OTFS channel estimation and signal detection, which are two important components of OTFS to attain its remarkable performance.

A. Related Work

Several OTFS channel estimation and signal detection algorithms have been developed in the literature. Subspace-aided OTFS scheme has been developed in [1] to handle Doppler-spread channel, which however requires a large number of dedicated pilots, leading to low spectral efficiency. Iterative OTFS receivers using superimposed pilots have been proposed in [2], [12], and [13]. A deep-learning-assisted OTFS channel estimation scheme has been developed by using superimposed pilots in [8]. However, these OTFS studies [2], [8], [12], [13] are suitable for Doppler-shift channel only. Meanwhile, the values of Doppler shifts need to be pre-estimated by using dedicated pilots in [2], reducing spectral efficiency. Two basis expansion modeling (BEM) OTFS receivers have been proposed for Doppler-spread channel in [14] and [15] by the authors, which are assisted with dedicated and superimposed pilots, respectively. However, channel coding, as a very important component in any communication systems, is not considered in the authors' related works [14], [15]. For instance, Turbo and convolutional coding have been adopted in the fourth generation (4G) long-term evolution (LTE) networks, while low-density parity-check (LDPC) and Polar coding have been utilized in the 5G new radio (NR) networks. Moreover, the aforementioned studies [1], [2], [12], [13], [14], [15] do not exploit *Turbo concept* to refine channel estimation, signal detection, and data decoding jointly for OTFS. *Turbo concept* has been adopted in [4], [16], [17], [18], [19], [20], and [21] to refine OTFS signal detection, while assuming perfect OTFS channel estimation. The authors in [22] and [23] have investigated iterative channel estimation, signal

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detection, and data decoding for single-carrier systems, which however are unapplicable to OTFS. Hence, the existing OTFS studies have the following limitations: a) Doppler-shift channel model [2], [8], [12], [13], b) impractical assumption of perfect channel knowledge [4], [16], [17], [18], [19], [20], [21], c) low spectral efficiency [1], d) limited reliability [14], [15], etc. To the best of the authors' knowledge, an iterative channel estimation, signal detection, and data decoding algorithm featuring high reliability and low pilot overhead is still an open problem in OTFS subject to Doppler-spread channel.

Like the works in [2], [8], [12], [13], and [15], superimposed pilots are adopted for OTFS channel estimation in this paper to achieve high spectral efficiency. Superimposed pilot power ratio is defined as the ratio of total transmission power used for pilot transmission. If superimposed pilot power ratio is small, initial channel estimation and signal detection by superimposed pilots may be worse and the subsequent refinement for channel estimation and signal detection may fail. Instead, if superimposed pilot power ratio is big, less power is remained for data transmission, leading to low reliability. Hence, it is of great importance to find a suitable superimposed pilot power ratio. The derivation of optimal superimposed pilot power ratio is not investigated in the authors' previous conference paper [15]. Besides, the other related OTFS studies [8], [13] did not set a transmission power constraint and the superimposed pilot power is supposed to be much higher than data power, leading to high power cost. The optimal superimposed pilot power ratio has been derived in [2], which however applies to Doppler-shift channel only. Furthermore, when using superimposed pilots, the selection of BEM order is also of great importance. The smaller BEM order tends to have a bigger modeling error, resulting in low reliability. The larger BEM order would require high pilot power and leaves less power for data transmission, yielding a decline in reliability. Hence, it is also crucial to choose a suitable BEM order. The authors believe that the optimization of superimposed pilot power ratio and BEM order is still an open problem in OTFS communications subject to Doppler-spread channel.

B. Contributions

In this paper, a Turbo BEM OTFS scheme has been proposed for high-mobility communications subject to either Doppler-shift or Doppler-spread channel. The proposed scheme includes superimposed pilot power ratio and BEM order optimization and Turbo BEM OTFS receiver with iterative channel estimation, signal detection, and data decoding. The contributions of this work are as follows:

- The authors believe that this is the first work optimizing superimposed pilot power ratio for OTFS communications subject to Doppler-spread channel, while the existing studies, such as, in [2], are applicable only to Doppler-shift channel. The signal-to-interference-and-noise-ratio (SINR) after channel estimation using superimposed pilots is derived as functions of OTFS system parameters and several channel information (e.g., channel correlation matrix, noise variance, etc.). The optimal superimposed pilot power ratio is obtained with a closed-form solution by calculating the first derivative

of derived SINR. BEM order is specially optimized for Doppler-spread channel, instead of simply increasing or reducing it. Simulation results validate the effectiveness of proposed Turbo BEM OTFS scheme for various channels. The derived SINR, optimized pilot power ratio and BEM order are consistent with their simulated ones.

- To the best of the authors' knowledge, this is the first solution employing *Turbo concept* to jointly refine channel estimation, signal detection and data decoding in OTFS communications. Channel estimation, signal detection, and data decoding are conducted in an iterative manner with exchange of soft information. In contrast, the previous works do not take into account channel coding into their systems, such as, [14] and [15]. Some of them [4], [16], [17], [18], [19], [20], [21] use soft information to refine signal detection only, while assuming perfect channel estimation. Simulation results show that the proposed scheme exhibits lower bit error rate (BER) than the existing schemes, such as [1] and [2].
- The proposed scheme uses superimposed pilots with optimized power for channel estimation, leading to higher spectral efficiency than dedicated-pilot aided OTFS schemes, such as [1]. Moreover, the proposed scheme does not need to transmit additional dedicated pilots to pre-estimate Doppler shifts, unlike [2], [12]. Spectral efficiency analysis validates the higher spectral efficiency of proposed scheme over existing schemes [1], [2]. Moreover, extrinsic information exchange transfer (EXIT) chart analysis and simulation results confirm the fast convergence speed of proposed scheme.

C. Organization and Notations

The rest of this paper is organized as follows: Section II describes the coded BEM OTFS system model subject to Doppler-spread channel. Section III describes the proposed Turbo BEM OTFS scheme. Performance analysis, including spectral efficiency analysis, EXIT chart analysis, and computational complexity analysis, are presented in Section IV. Simulation results are provided in Section V. The paper is closed in Section VI with conclusions.

Notations: Bold symbols represent vectors/matrices. Superscripts T , H , and $\dagger$ denote the transpose, complex conjugate transpose, and pseudo inverse of a vector/matrix, respectively. $\text{diag}\{\mathbf{a}\}$ is a diagonal matrix with vector $\mathbf{a}$ on its diagonal. $\otimes$ indicates the Kronecker product.

II. SYSTEM MODEL

Fig. 1 illustrates the block diagram of coded OTFS system. Denote $\mathbf{b}$ as bit stream which is then encoded by channel encoder with code rate of r , yielding the coded bit stream $\mathbf{b}_c$. $\mathbf{b}_c$ is first interleaved and then mapped to data symbols as $\mathbf{x}_d$ using phase shift keying (PSK) or quadrature amplitude modulation (QAM) modulation. Superimposed pilots denoted as $\mathbf{x}_{sp}$ are adopted and added on top of every data symbol $\mathbf{x}_d$. Note that $\mathbf{x}_d$ and $\mathbf{x}_{sp}$ are column vectors of length MN , where M and N are respectively the number of OTFS delay and Doppler grids/bins. Define α ($\alpha \in [0, 1]$) as superimposed pilot power ratio. The transmitted OTFS signal vector of length

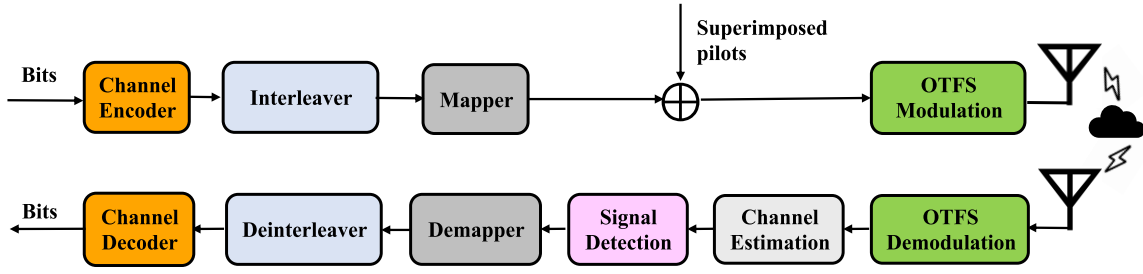


Fig. 1. Block diagram of coded OTFS system.

MN in delay-Doppler domain $\mathbf{x}$ is expressed as

$$\mathbf{x} = \sqrt{\alpha} \mathbf{x}_{\text{sp}} + \sqrt{1 - \alpha} \mathbf{x}_{\text{d}}. \quad (1)$$

Delay-Doppler channel representation has been considered in many OTFS papers, such as [2] and [24]. It is deterministic with each path associated with a delay and a Doppler shift. This deterministic channel representation is useful to some extent, but may not be feasible to characteristic time-varying channels in all scenarios of practical relevance. To cater for a wide variety of communication scenarios, stochastic channel model, with some kind of randomness and uncertainty, is introduced in Section 1.4 of [25]. The stochastic representation of a time-varying channel corresponds to its second-order statistics, *i.e.*, channel autocorrelation in time domain or Doppler spectrum in frequency domain. The stochastic channel with Jakes Doppler spectrum has been recommended in many standards for outdoor scenarios, such as 5G NR [26]. The autocorrelation function of Jakes spectrum is expressed as $J_0(2\pi f_{\text{max}} T_s \Delta t)$, where $J_0(\cdot)$ is the Bessel function of first kind of order zero, T_s sampling time, f_{max} maximum Doppler frequency as $f_{\text{max}} = \frac{f_c v}{c}$, with f_c , v , and c being respectively carrier frequency, velocity, and speed of light [25].

Taking one channel path as an example, Fig. 2 shows the time-varying channel gain, channel autocorrelation, and Doppler spectrum of delay-Doppler channel and stochastic channel with Jakes Doppler spectrum, respectively. Their differences are summarized and highlighted as follows: a) the channel gain of stochastic channel changes more randomly and dramatically over time than that of delay-Doppler channel, b) stochastic channel becomes less correlated over time, and c) stochastic channel has a wide Doppler spread comprising multiple Doppler shifts rather than a single Doppler shift. Therefore, delay-Doppler channel and stochastic channel are referred to as Doppler-shift channel and Doppler-spread channel in this paper. Due to the aforementioned features of Doppler-spread channels, more randomness and uncertainties, less correlation, wide Doppler-spread, delay-Doppler channel representation with each path associated with a delay and a Doppler shift may not be suitable for Doppler-spread channels. BEM is thus introduced to parametric Doppler-spread channel as a weighted combination of basis functions.

Wide sense stationary channel is considered in this paper, which implies that the second-order statistics of channel, *i.e.*, channel autocorrelation function and Doppler spectrum, are stationary over some time [25]. From a perspective of practical scenarios, it is reasonable to have a stationary Doppler spectrum over tens of OTFS frames. The OTFS frame length is usually 1 ms [24] and the velocity (maximum Doppler

frequency) and environment (Doppler spectrum type) do not change too much during this short period of time. Similarly to existing OTFS papers, *e.g.*, [24], maximum Doppler frequency should be smaller than OTFS subcarrier spacing. The Doppler spectrum type would not affect OTFS frame design. Doubly-selective fading channel with $(L + 1)$ paths is considered. Assume that each channel path has a single Doppler shift [2], [24] or Doppler spread consisting of multiple Doppler shifts [1], [14]. Denote $h[t, l']$ as channel gain of l' -th path and t -th time instant, with $l' = 0, 1, \dots, L$ and $t = 0, 1, \dots, MN - 1$. By applying BEM, $h[t, l']$ is written as

$$h[t, l'] = \sum_{q=0}^Q c_q[l'] e^{\frac{j2\pi(q-Q/2)t}{MNR}} + e[t, l'], \quad (2)$$

where $c_q[l']$ is q -th BEM coefficient of l' -th channel path, Q BEM order, R BEM resolution parameter, and $e[t, l']$ BEM modeling error. R is typically given by 1 for integer Doppler-shift channel, and 2 for fractional Doppler-shift and Doppler-spread channel. The detailed discussion on the choice of R can refer to the work in [14]. Denote $\mathbf{h}_{l'} = [h[0, l'], h[1, l'], \dots, h[MN - 1, l']]^T$ and $\mathbf{e}_{l'} = [e[0, l'], e[1, l'], \dots, e[MN - 1, l']]^T$. $\mathbf{h}_{l'}$ can be expressed as

$$\mathbf{h}_{l'} = \sum_{q=0}^Q \mathbf{b}_q c_q[l'] + \mathbf{e}_{l'}, \quad (3)$$

where the q -th BEM basis function used for channel modeling is defined as $\mathbf{b}_q = [1, e^{\frac{j2\pi(q-Q/2)}{MNR}}, \dots, e^{\frac{j2\pi(q-Q/2)(MN-1)}{MNR}}]$. According to [14], the mean-square-error (MSE) of BEM modeling is given by

$$\text{MSE}_{\text{mod}} = \frac{\sum_{l'=0}^L \text{Tr}\{(\mathbf{I}_{MN} - \mathbf{B}(\mathbf{B}^H \mathbf{B})^{-1} \mathbf{B}^H) \mathbf{R}_{\text{hh}, l'}\}}{MN(L + 1)}, \quad (4)$$

where $\mathbf{R}_{\text{hh}, l'}$ is channel correlation matrix of l' -th path and $\mathbf{B} = [\mathbf{b}_0, \mathbf{b}_1, \dots, \mathbf{b}_Q]$. Fig. 4 provides an example for BEM basis functions with $Q = 4$. It can be seen that each BEM basis $\mathbf{b}_q$ is deterministic and varying over time, while its corresponding BEM coefficient c_q remains unchanged over one OTFS frame and needs to be estimated using pilots. Note that the type of BEM used in this paper is generalized complex exponential BEM. Other BEM models, such as, polynomial BEM and discrete prolate spheroidal BEM in [25], can also be applied with proposed OTFS scheme.

In addition, BEM has a wide application in high-mobility orthogonal frequency division multiplexing (OFDM) communications and more details can refer to Chapters 1, 4 and 7 of [25]. Though the existing BEM OFDM receivers could

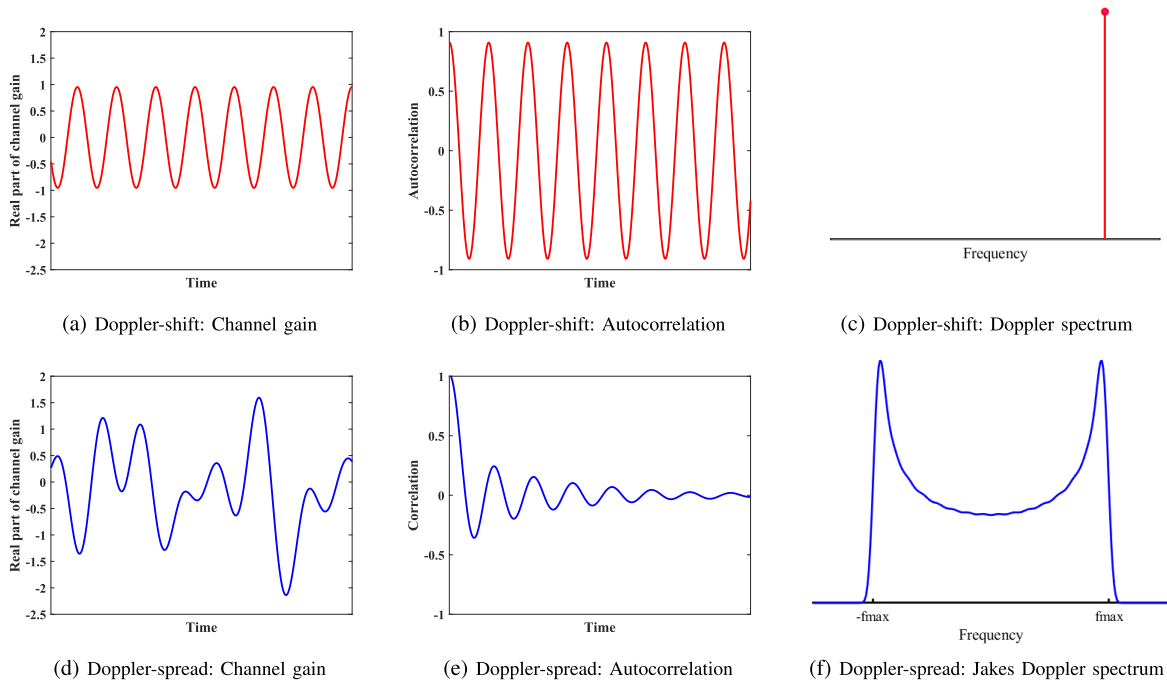


Fig. 2. Comparison of Doppler-shift and Doppler-spread channel in terms of time-varying channel gain, channel autocorrelation, and Doppler spectrum.

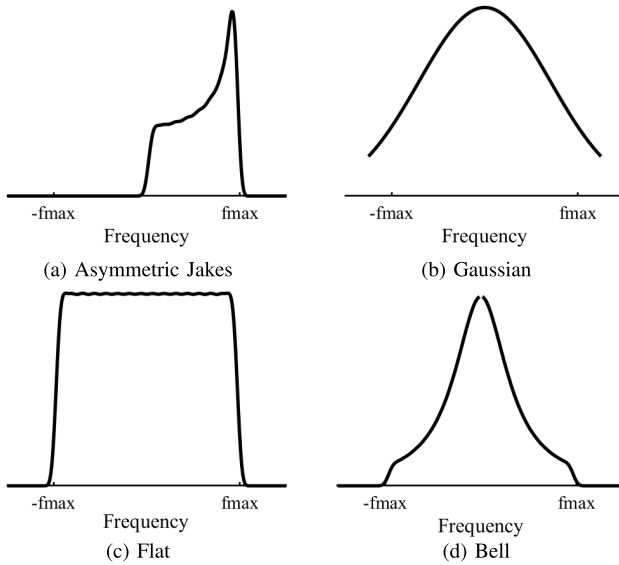


Fig. 3. Other shapes of Doppler spectrum.

work for Doppler-spread channel [25], OFDM communications has an inherent disadvantage compared to OTFS communications. According to [27] and [28], OFDM communications has a lower diversity order than OTFS communications in doubly selective fading channels. Hence, to enhance OTFS communication and preserve its superiority over OFDM, it is necessary to investigate BEM OTFS receivers especially in high-mobility communications with Doppler-spread channel.

In addition to the single-Doppler-shift spectrum and Jakes spectrum in Fig. 2, there are other types of Doppler spectrum in Fig. 3, namely asymmetric Jakes, Gaussian, flat, bell, etc. Each Doppler spectrum has its specific use cases. Jakes spectrum, also known as classical spectrum, is recommended by most standards for outdoor environments [26].

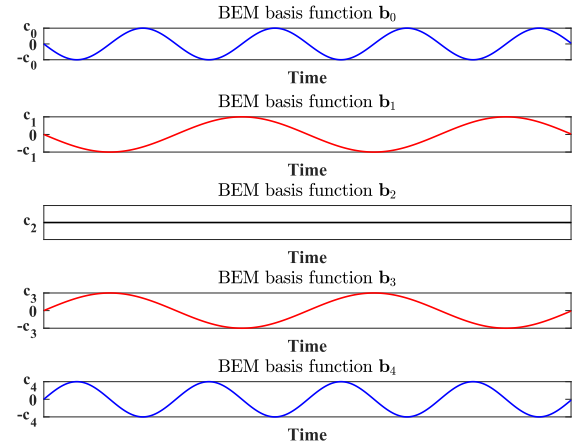


Fig. 4. Example of BEM basis functions. BEM basis functions are deterministic and time-varying, each with an unknown time-invariant BEM coefficient to be estimated using pilots.

Its assumption of uniform angle distribution of scatterers makes it unapplicable to microcells in densely populated urban areas. Asymmetric Jakes spectrum happens in vehicle-to-vehicle (V2V) communications, when two terminals move in the opposite direction [29]. Jakes and asymmetric Jakes spectrums have peak power near $f_{\max}$ and their power diminishes elsewhere. In contrast, the power of Gaussian-shaped and bell-shaped spectrums, is concentrated in the middle and decays as frequency goes toward $f_{\max}$. Gaussian-shaped and bell-shaped spectrum are usually for indoor environments with fixed terminals and moving scatterers [30]. Flat Doppler spectrum is also for indoor environments but with fixed scatters, and its power is flat over Doppler frequency range [31]. Although indoor environments have small velocity v , Doppler frequency may still be significant when high-frequency bands are adopted.

Denote $\mathbf{T} = \mathbf{F}_N \otimes \mathbf{I}_M$, with $\mathbf{F}_N$ and $\mathbf{I}_M$ being N -point discrete Fourier transform (DFT) matrix and $M \times M$ identity

matrix, respectively. After passing through OTFS modulation, doubly selective fading channel, and OTFS demodulation, following the derivation in [14], the received BEM OTFS signal in delay-Doppler domain can be expressed as

$$\mathbf{y} = \sum_{q=0}^Q \mathbf{T} \text{diag}\{\mathbf{b}_q\} \mathbf{F}_{MN}^H \text{diag}\{\mathbf{F}_{MN \times L} \mathbf{c}_q\} \mathbf{F}_{MN} \mathbf{T}^H \mathbf{x} + \mathbf{w} + \mathbf{z}, \quad (5)$$

where $\mathbf{F}_{MN}$ is MN -point DFT matrix, $\mathbf{c}_q = [c_q[0], c_q[1], \dots, c_q[L]]^T$ is BEM coefficient vector, $\mathbf{w}$ is the additive white Gaussian noise (AWGN) vector of variance $\sigma_w^2 \mathbf{I}_{MN}$, and $\mathbf{z}$ is the error to the received signal caused by BEM modeling. According to [14], $\mathbf{z}$ can be considered as AWGN vector with zero mean and variance of $\sigma_z^2 \mathbf{I}_{MN}$. Note that (5) is equivalent to

$$\mathbf{y} = \sqrt{\alpha} \sum_{q=0}^Q \mathbf{A}_{\text{sp},q} \mathbf{c}_q + \sqrt{1-\alpha} \sum_{q=0}^Q \mathbf{A}_{\text{d},q} \mathbf{c}_q + \mathbf{w} + \mathbf{z}, \quad (6)$$

with $\mathbf{A}_{\text{sp},q} = \mathbf{T} \text{diag}\{\mathbf{b}_q\} \mathbf{F}_{MN}^H \text{diag}\{\mathbf{F}_{MN} \mathbf{T}^H \mathbf{x}_{\text{sp}}\} \mathbf{F}_{MN \times L}$ and $\mathbf{A}_{\text{d},q} = \mathbf{T} \text{diag}\{\mathbf{b}_q\} \mathbf{F}_{MN}^H \text{diag}\{\mathbf{F}_{MN} \mathbf{T}^H \mathbf{x}_{\text{d}}\} \mathbf{F}_{MN \times L}$. The detailed derivations for (5) and (6) can refer to [14].

The research focuses of this paper are to: a) optimize superimposed pilot power ratio α and BEM order Q , and b) design a Turbo BEM OTFS receiver. Superimposed pilot power ratio has been optimized in [2], which however is for Doppler-shift channel only. Moreover, several Turbo OTFS receivers [4], [16], [17], [18], [19], [20], [21] have been developed in the literature, which however assume perfect channel estimation. To further enhance reliability, this paper aims to design a Turbo BEM OTFS receiver with soft information exchange between BEM channel estimation, signal detection, and data decoding.

III. TURBO BEM OTFS SCHEME

This section aims to describe the proposed Turbo BEM OTFS scheme. Superimposed pilot power ratio and BEM order are first optimized to maximize the derived SINR which is as functions of several OTFS system and channel parameters. Turbo BEM OTFS receiver with iterative BEM channel estimation, signal detection, and data decoding is then proposed. In the beginning, superimposed pilots with optimized power ratio are utilized to perform an initial (rough) BEM channel estimation using the optimized BEM order, while data is treated as interference. Then, signal detection, demapping, deinterleaver, and channel decoding are implemented to generate *a posteriori* and extrinsic log-likelihood ratios (LLRs) of data bits, which are then fed back to refine BEM channel estimation, signal detection, and data decoding iteratively.

A. Pilot Power Ratio and BEM Order Optimization

In this subsection, the optimization algorithm for superimposed pilot power ratio and BEM order is described for OTFS communications subject to Doppler-shift [2], [24] or Doppler-spread channel [1], [14]. According to [2], the optimal superimposed pilot power ratio can be obtained by maximizing SINR after channel estimation using superimposed pilots. Thus, this subsection is focused on deriving the SINR as functions of BEM order and superimposed pilot power ratio after pilot-aided BEM channel estimation.

1) *Derivation of SINR*: The received OTFS signal in delay-Doppler domain in (6) can be rewritten as

$$\mathbf{y} = \sqrt{\alpha} \mathbf{A}_{\text{sp}} \mathbf{c} + \underbrace{\sqrt{1-\alpha} \mathbf{A}_{\text{d}} \mathbf{c} + \mathbf{w} + \mathbf{z}}_{\text{Data interference + Noise + Modeling error}}, \quad (7)$$

with $\mathbf{c} = [\mathbf{c}_0^T, \mathbf{c}_1^T, \dots, \mathbf{c}_Q^T]^T$, $\mathbf{A}_{\text{sp}} = [\mathbf{A}_{\text{sp},0}, \mathbf{A}_{\text{sp},1}, \dots, \mathbf{A}_{\text{sp},Q}]$, and $\mathbf{A}_{\text{d}} = [\mathbf{A}_{\text{d},0}, \mathbf{A}_{\text{d},1}, \dots, \mathbf{A}_{\text{d},Q}]$. By treating data as interference, initial BEM coefficient vector can be obtained by utilizing superimposed pilot matrix $\mathbf{A}_{\text{sp}}$ and denoted as $\hat{\mathbf{c}}^0$. Hence, the received data signal is given by

$$\begin{aligned} \hat{\mathbf{y}}_{\text{d}}^0 &= \mathbf{y} - \sqrt{\alpha} \mathbf{A}_{\text{sp}} \hat{\mathbf{c}}^0 \\ &= \sqrt{\alpha} \mathbf{A}_{\text{sp}} \mathbf{c} + \sqrt{1-\alpha} \mathbf{A}_{\text{d}} \mathbf{c} + \mathbf{w} + \mathbf{z} - \sqrt{\alpha} \mathbf{A}_{\text{sp}} \hat{\mathbf{c}}^0 \\ &= \underbrace{\sqrt{1-\alpha} \mathbf{A}_{\text{d}} \hat{\mathbf{c}}^0}_{\text{Desired signal}} + \underbrace{\mathbf{A}(\mathbf{c} - \hat{\mathbf{c}}^0) + \mathbf{w} + \mathbf{z}}_{\text{Estimation error + Noise + Modeling error}}, \end{aligned} \quad (8)$$

with $\mathbf{A} = \sqrt{\alpha} \mathbf{A}_{\text{sp}} + \sqrt{1-\alpha} \mathbf{A}_{\text{d}}$. The SINR after superimposed-pilot aided BEM channel estimation is defined as

$$\text{SINR}_0(Q, \alpha) = \frac{\mathbb{E}\{(1-\alpha)(\hat{\mathbf{c}}^0)^H \mathbf{A}_{\text{d}}^H \mathbf{A}_{\text{d}} \hat{\mathbf{c}}^0\}}{\mathbb{E}\{(\tilde{\mathbf{c}}^0)^H \mathbf{A}^H \mathbf{A} \tilde{\mathbf{c}}^0 + \mathbf{w}^H \mathbf{w} + \mathbf{z}^H \mathbf{z}\}}, \quad (9)$$

with $\tilde{\mathbf{c}}^0 = \mathbf{c} - \hat{\mathbf{c}}^0$. Denote $\text{SINR}_{0,\text{D}}$ and $\text{SINR}_{0,\text{N}}$ as the denominator and numerator of (9), respectively. $\text{SINR}_{0,\text{D}}$ comprises the noise and interference due to estimation error and modeling error, while $\text{SINR}_{0,\text{N}}$ is the desired data signal power. $\text{SINR}_{0,\text{D}}$ and $\text{SINR}_{0,\text{N}}$ are derived below, respectively.

Because $\mathbf{w}$ and $\mathbf{z}$ are an Gaussian random vector with variance of $\sigma_w^2 \mathbf{I}_{MN}$ and $\sigma_z^2 \mathbf{I}_{MN}$, $\text{SINR}_{0,\text{D}}$ is reexpressed as

$$\text{SINR}_{0,\text{D}} = \mathbb{E}\{\text{Tr}\{\mathbf{A}^H \mathbf{A} \tilde{\mathbf{c}}^0 (\tilde{\mathbf{c}}^0)^H\}\} + MN(\sigma_w^2 + \sigma_z^2). \quad (10)$$

According to [14], the MSE of BEM coefficient estimation by using superimposed pilots is

$$\mathbb{E}\{\tilde{\mathbf{c}}^0 (\tilde{\mathbf{c}}^0)^H\} = (\mathbf{A}_{\text{sp}}^H \mathbf{A}_{\text{sp}})^{-1} \frac{\sigma_w^2 + \sigma_z^2 + 1 - \alpha}{\alpha}. \quad (11)$$

The first item of (10) is set to K , which is further expressed as (12), shown at the bottom of the next page, with $s = \text{Tr}\{\mathbf{A}_{\text{d}}^H \mathbf{A}_{\text{d}} (\mathbf{A}_{\text{sp}}^H \mathbf{A}_{\text{sp}})^{-1}\}$. Therefore, $\text{SINR}_{0,\text{D}}$ is finally expressed as (13), shown at the bottom of the next page. Because of $\mathbf{c} = \hat{\mathbf{c}}^0 + \tilde{\mathbf{c}}^0$, the BEM coefficient estimate $\hat{\mathbf{c}}^0$ is expressed as $\hat{\mathbf{c}}^0 = \mathbf{c} - \tilde{\mathbf{c}}^0$, and its correlation matrix is calculated as

$$\begin{aligned} \mathbb{E}\{\hat{\mathbf{c}}^0 (\hat{\mathbf{c}}^0)^H\} &= \mathbb{E}\{(\mathbf{c} - \tilde{\mathbf{c}}^0)(\mathbf{c}^H - (\tilde{\mathbf{c}}^0)^H)\} \\ &= \mathbb{E}\{\tilde{\mathbf{c}}^0 (\tilde{\mathbf{c}}^0)^H\} + \mathbb{E}\{\mathbf{c} \mathbf{c}^H\} \\ &= (\mathbf{A}_{\text{sp}}^H \mathbf{A}_{\text{sp}})^{-1} \frac{\sigma_w^2 + \sigma_z^2 + 1 - \alpha}{\alpha} + \mathbb{E}\{\mathbf{c} \mathbf{c}^H\}. \end{aligned} \quad (14)$$

As a result, $\text{SINR}_{0,\text{N}}$ can be expressed as

$$\begin{aligned} \text{SINR}_{0,\text{N}} &= \mathbb{E}\{\text{Tr}\{(1-\alpha) \mathbf{A}_{\text{d}}^H \mathbf{A}_{\text{d}} (\tilde{\mathbf{c}}^0 (\tilde{\mathbf{c}}^0)^H + \mathbf{c} \mathbf{c}^H)\}\} \\ &= \text{Tr}\{(1-\alpha)(\mathbb{E}\{\mathbf{A}_{\text{d}}^H \mathbf{A}_{\text{d}} \tilde{\mathbf{c}}^0 (\tilde{\mathbf{c}}^0)^H\} + \mathbb{E}\{\mathbf{A}_{\text{d}}^H \mathbf{A}_{\text{d}} \mathbf{c} \mathbf{c}^H\})\} \\ &= (1-\alpha) \left(\frac{s(\sigma_w^2 + \sigma_z^2 + 1 - \alpha)}{\alpha} + \text{Tr}\{\mathbb{E}\{\mathbf{A}_{\text{d}}^H \mathbf{A}_{\text{d}} \mathbf{c} \mathbf{c}^H\}\} \right). \end{aligned} \quad (15)$$

Following the derivation in Appendix A, $\text{Tr}\{\mathbb{E}\{\mathbf{A}_d^H \mathbf{A}_d \mathbf{c} \mathbf{c}^H\}\}$ in (15) is derived as $\text{Tr}\{\mathbf{R}_{hh}\}$, where $\mathbf{R}_{hh}$ is channel correlation matrix. Denote $p = \text{Tr}\{\mathbf{R}_{hh}\}$, $\text{SINR}_{0,N}$ is finally given by

$$\text{SINR}_{0,N} = \frac{(1 - \alpha)[s(\sigma_w^2 + \sigma_z^2 + 1 - \alpha) + p\alpha]}{\alpha}. \quad (16)$$

By substituting (13) and (16) to (9), the SINR after initial pilot-aided BEM channel estimation is derived as (17), shown at the bottom of the page. Assuming full rank of $\mathbf{A}_d^H \mathbf{A}_d$ and $\mathbf{A}_{sp}^H \mathbf{A}_{sp}$ are achieved by using proper BEM modeling, s can be approximated as the full rank of $\mathbf{A}_d^H \mathbf{A}_d (\mathbf{A}_{sp}^H \mathbf{A}_{sp})^{-1}$ and calculated as $(L + 1)(Q + 1)$.

2) *Optimization of Pilot Power Ratio and BEM Order:* The superimposed pilot power ratio α and BEM order Q can be optimized by maximizing the derived SINR in (17),

$$\{Q^*, \alpha^*\} = \max_{Q, \alpha} \text{SINR}_0(Q, \alpha). \quad (18)$$

However, it is difficult to find the direct solutions to (18) with two variables. (18) is thus decomposed into two subproblems:

$$\alpha_Q^* = \max_{\alpha} \text{SINR}_0(Q, \alpha), \quad (19a)$$

$$Q^* = \max_Q \text{SINR}_0(Q, \alpha_Q^*), \quad (19b)$$

where the search range of BEM order is $Q \geq 1$. Hence, the optimal pilot power ratio is obtained first for a certain value of Q . Denote $b_1 = s - p$, $b_2 = p - 2s - s(\sigma_w^2 + \sigma_z^2)$, $b_3 = 1 + \sigma_w^2 + \sigma_z^2$, and $\sigma_{wz}^2 = \sigma_w^2 + \sigma_z^2$. BEM modeling error σ_z^2 is dependent on BEM order Q , and its value is determined by using (4). To solve (19a), the first derivative of $\text{SINR}_0(Q, \alpha)$ with respect to α is derived and the optimal α_Q^* is obtained by solving the function below,

$$\begin{aligned} \frac{d \text{SINR}_0(Q, \alpha)}{d \alpha} &= \frac{d \frac{b_1 \alpha^2 + b_2 \alpha + b_3}{(MN\sigma_{wz}^2 - s)\alpha + sb_3}}{d \alpha} \\ &= \frac{d_1 \alpha^2 + d_2 \alpha + d_3}{((MN\sigma_{wz}^2 - s)\alpha + sb_3)^2} = 0 \end{aligned} \quad (20)$$

with $d_1 = b_1 MN\sigma_{wz}^2 - b_1 s$, $d_2 = 2b_1 b_3 s$, and $d_3 = b_2 b_3 s - b_3 MN\sigma_{wz}^2 + b_3 s^2$. Two solutions to (20) are obtained as

$$\hat{\alpha}_{Q,1} = \frac{-d_2 + \sqrt{d_2^2 - 4d_1 d_3}}{2d_1}, \quad \hat{\alpha}_{Q,2} = \frac{-d_2 - \sqrt{d_2^2 - 4d_1 d_3}}{2d_1}. \quad (21)$$

Define j as the j -th ($j = 1, 2$) solution of α . Because of $\alpha \in [0, 1]$, the optimal superimposed pilot power ratio α_Q^* is

$$\alpha_Q^* = \hat{\alpha}_{Q,j}, \text{ if } \hat{\alpha}_{Q,j} \in (0, 1). \quad (22)$$

The optimization starts with $Q = 1$ in (19a), and then repeat the calculation in (19a) with $Q = Q + 1$ each time, until $\text{SINR}_0(Q, \alpha_Q^*) < \text{SINR}_0(Q - 1, \alpha_{Q-1}^*)$. The optimal BEM order and pilot power ratio are then given by $Q^* = Q - 1$ and $\alpha^* = \alpha_{Q-1}^*$, which will be used with Turbo BEM OTFS receiver in Subsection III-B. The proposed pilot power ratio and BEM order optimization is summarized in Algorithm 1.

Algorithm 1: Optimization of Superimposed Pilot Power Ratio and BEM order

Input: Number of OTFS delay grids M , number of OTFS Doppler grids N , channel correlation matrix $\mathbf{R}_{hh}$, maximum channel length L , and Gaussian noise variance σ_w^2 .

Output: Optimal BEM order Q^* and optimal superimposed pilot power ratio α^* .

- 1 Initialization: BEM order $Q = 1$
 - 2 **repeat**
 - 3 Obtaining BEM modeling error by using (4) with BEM order Q ,
 - 4 Finding its optimal superimposed pilot power ratio α_Q^* and its corresponding $\text{SINR}_0(Q, \alpha_Q^*)$ by using (17)-(22),
 - 5 $Q = Q + 1$.
 - 6 **until** $\text{SINR}_0(Q, \alpha_Q^*) < \text{SINR}_0(Q - 1, \alpha_{Q-1}^*)$;
 - 7 **return:** $Q^* = Q - 1$ and $\alpha^* = \alpha_{Q-1}^*$.
-

According to Algorithm 1, in addition to OTFS system parameters, several channel-related parameters need to be

$$\begin{aligned} K &= \mathbb{E}\{\text{Tr}\{(\alpha \mathbf{A}_{sp}^H \mathbf{A}_{sp} + (1 - \alpha) \mathbf{A}_d^H \mathbf{A}_d)(\mathbf{A}_{sp}^H \mathbf{A}_{sp})^{-1} \frac{\sigma_w^2 + \sigma_z^2 + 1 - \alpha}{\alpha}\}\} \\ &= (L + 1)(Q + 1) \alpha \frac{\sigma_w^2 + \sigma_z^2 + 1 - \alpha}{\alpha} + \text{Tr}\{(1 - \alpha) \mathbf{A}_d^H \mathbf{A}_d (\mathbf{A}_{sp}^H \mathbf{A}_{sp})^{-1} \frac{\sigma_w^2 + \sigma_z^2 + 1 - \alpha}{\alpha}\} \\ &= \frac{(L + 1)(Q + 1)(\sigma_w^2 + \sigma_z^2 + 1 - \alpha)\alpha + (1 - \alpha)(\sigma_w^2 + \sigma_z^2 + 1 - \alpha)s}{\alpha}. \end{aligned} \quad (12)$$

$$\text{SINR}_{0,D} = \frac{(\sigma_w^2 + \sigma_z^2 + 1 - \alpha)((1 - \alpha)s + \alpha(L + 1)(Q + 1)) + MN\alpha(\sigma_w^2 + \sigma_z^2)}{\alpha}. \quad (13)$$

$$\begin{aligned} \text{SINR}_0(Q, \alpha) &= \frac{(1 - \alpha)[s(\sigma_w^2 + \sigma_z^2 + 1 - \alpha) + p\alpha]}{(s - p)\alpha^2 + (p - 2s - s(\sigma_w^2 + \sigma_z^2))\alpha + s(\sigma_w^2 + \sigma_z^2 + 1)} \\ &= \frac{(1 - \alpha)[s(\sigma_w^2 + \sigma_z^2 + 1 - \alpha) + p\alpha]}{(s - p)\alpha^2 + (p - 2s - s(\sigma_w^2 + \sigma_z^2))\alpha + s(\sigma_w^2 + \sigma_z^2 + 1)} + \frac{MN\alpha(\sigma_w^2 + \sigma_z^2)}{(s - p)\alpha^2 + (p - 2s - s(\sigma_w^2 + \sigma_z^2))\alpha + s(\sigma_w^2 + \sigma_z^2 + 1)}. \end{aligned} \quad (17)$$

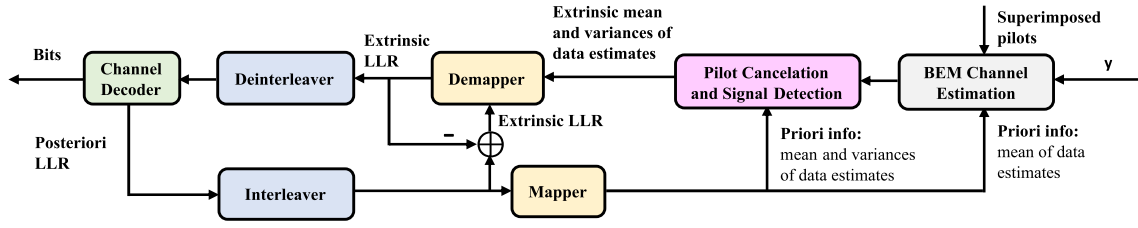


Fig. 5. Block diagram of Turbo BEM OTFS receiver with soft information exchange between BEM channel estimation, signal detection, and data decoding.

known for optimization, such as, channel correlation matrix $\mathbf{R}_{hh}$, channel length L , and noise variance σ_w^2 . Note that $\mathbf{R}_{hh}$ and L change slowly compared to instantaneous channel gain h_l , and are assumed to be acquired using existing methods (e.g., [32], [33]) before performing the proposed scheme. Moreover, the derived expression is as a function of channel correlation matrix $\mathbf{R}_{hh}$ and there is no specific assumption on $\mathbf{R}_{hh}$. Hence, the proposed optimization algorithm considering superimposed pilot-aided BEM channel estimation could accommodate to various channel model, such as, Doppler-shift channel with known/unknown Doppler shifts in [2], [12], and [13] and Doppler-spread channel in [1], [14], and [15]. According to [2], Doppler shifts are pre-estimated using dedicated pilots in its 1st OTFS frame. Hence, the derived superimposed pilot power ratio in [2] for channel-gain estimation may not be suitable for estimating both Doppler shifts and channel gains using superimposed pilots. The proposed scheme does not need additional dedicated pilots to pre-estimate Doppler shifts. Only superimposed pilots are used to estimate both Doppler shifts and channel gains implicitly in BEM channel estimation. Hence, the derived superimposed pilot power ratio of proposed scheme is suitable for estimating both Doppler shifts and channel gains of Doppler-shift channel. Note that one challenge of considering Doppler-spread channel lies in the derivation of initial SINR of vector-wise BEM OTFS system model. The initial SINR expression with cumbersome matrix manipulations in (9) is finally derived as simple multiplications and additions of scalars in (17). Hence, pilot power ratio optimization problem can be simplified by calculating the first derivative of initial SINR. Another challenge of Doppler-spread channel is the selection of a suitable BEM order Q . Simply increasing Q would not enhance performance. Instead, it could worsen performance. BEM order optimization is thus proposed in this subsection to find a proper BEM order.

B. Turbo BEM OTFS Receiver

This subsection aims to describe the proposed Turbo BEM OTFS receiver including iterative BEM channel estimation with optimized BEM order, pilot cancellation and signal detection, and data decoding. In the initial stage, channel is estimated by using superimposed pilots with optimized power ratio, while treating data as interference, which is then followed by pilot cancellation, signal detection, demapping, deinterleaver, and channel decoding. In the iterative stage, *a posteriori* LLRs from channel decoder are fed back to reconstruct *a posteriori* mean and variance of data symbol, which then refine BEM channel estimation and signal detection. Its

extrinsic LLRs from channel decoder are used as *a priori* LLRs of demapper. The above procedures between BEM channel estimation, signal detection, and data decoding are repeated until the stopping criterion are satisfied. In the following, BEM channel estimation, signal detection, demapping, channel decoding, and mapping are described, respectively.

1) *BEM Channel Estimation*: Initial BEM channel estimation is performed using superimposed pilots only, while iterative BEM channel estimation is implemented by using both superimposed pilots and fed-back mean of data estimates. Initial and iterative BEM channel estimation are described, respectively, in the following.

Initial BEM Channel Estimation: By applying the optimal superimposed power ratio α^* derived in Subsection III-A and treating data as interference, the received OTFS signal is

$$\mathbf{y} = \sqrt{\alpha^*} \mathbf{A}_{sp} \mathbf{c} + \underbrace{\sqrt{1 - \alpha^*} \mathbf{A}_d \mathbf{c} + \mathbf{w} + \mathbf{z}}_{\text{Data interference + Noise + Modeling error}}. \quad (23)$$

Hence, the BEM coefficient vector $\mathbf{c}$ can be initially estimated using least square algorithm [34] as

$$\hat{\mathbf{c}}^0 = \mathbf{A}_{sp}^\dagger \mathbf{y} / \sqrt{\alpha^*}. \quad (24)$$

Note that the superscript 0 indicates this is initial estimation.

Iterative BEM Channel Estimation: In addition to superimposed pilots, the *a priori* information, i.e., *a priori* mean of data symbol of $(i-1)$ -th iteration $\mathbf{m}_{pri}^{i-1}$, is fed back to refine channel estimation, obtaining BEM coefficient $\hat{\mathbf{c}}^i$. Note that $\mathbf{m}_{pri}^{i-1}$ is obtained using (38). To avoid noise enhancement, the power of $\mathbf{m}_{pri}^{i-1}$ is normalized with respect to its power for PSK modulation. For QAM modulation, it is normalized to the power of the symbol χ_h from QAM modulation alphabet $\mathbb{K}_{QAM}$, whose symbol probability $P_{\text{pos}, \mathbf{x}_d[n] = \chi_h}^{i-1}$ is highest among all candidates in QAM modulation alphabet. Define the normalized mean of data symbol estimate as $\bar{\mathbf{m}}_{pri}^{i-1}$. According to (6), $\hat{\mathbf{A}}_{d,q}^{i-1}$ can be formulated as

$$\hat{\mathbf{A}}_{d,q}^{i-1} = \mathbf{T} \text{diag}\{\mathbf{b}_q\} \mathbf{F}_{MN}^H \text{diag}\{\mathbf{F}_{MN} \mathbf{T}^H \bar{\mathbf{m}}_{pri}^{i-1}\} \mathbf{F}_{MN \times L}. \quad (25)$$

$\hat{\mathbf{A}}_d^{i-1}$ is obtained as $\hat{\mathbf{A}}_d^{i-1} = [\hat{\mathbf{A}}_{d,0}^{i-1}, \hat{\mathbf{A}}_{d,1}^{i-1}, \dots, \hat{\mathbf{A}}_{d,Q}^{i-1}]$. The BEM coefficient vector $\mathbf{c}$ of i -th iteration is refined using least square algorithm [34] as

$$\hat{\mathbf{c}}^i = (\hat{\mathbf{A}}^{i-1})^\dagger \mathbf{y}, \quad (26)$$

with $\hat{\mathbf{A}}^{i-1} = (\sqrt{\alpha^*} \mathbf{A}_{sp} + \sqrt{1 - \alpha^*} \hat{\mathbf{A}}_d^{i-1})$.

2) *Pilot Cancellation and Signal Detection*: With the BEM coefficient vector estimate from i -th iteration $\hat{\mathbf{c}}^i$, the received data signal can be determined by subtracting the pilot signal from the received signal, *i.e.*,

$$\begin{aligned}\hat{\mathbf{y}}_d^i &= \mathbf{y} - \sqrt{\alpha^*} \mathbf{A}_{sp} \hat{\mathbf{c}}^i \\ &= \sqrt{1 - \alpha^*} \mathbf{A}_d \mathbf{c} + \sqrt{\alpha^*} \mathbf{A}_{sp} (\mathbf{c} - \hat{\mathbf{c}}^i) + \mathbf{w} + \mathbf{z},\end{aligned}\quad (27)$$

Denote $\mathbf{e}^i = \sqrt{\alpha^*} \mathbf{A}_{sp} (\mathbf{c} - \hat{\mathbf{c}}^i)$. $\mathbf{e}^i$ is error due to channel estimation and can be regarded as a Gaussian random vector with zero mean and variance of $\sigma_{e,i}^2$. $\sigma_{e,i}^2$ is expressed as

$$\sigma_{e,i}^2 = \begin{cases} \frac{(L+1)(Q+1)(\sigma_w^2 + \sigma_z^2 + 1 - \alpha^*)}{MN} & i = 0 \\ \beta & i \geq 1 \end{cases} \quad (28)$$

where $\beta = \text{Tr}\{\alpha^* \mathbf{A}_{sp} ((\mathbf{A}_d^i)^H \mathbf{A}_d^i)^{-1} (\sigma_w^2 + \sigma_z^2) \mathbf{A}_{sp}^H\} / (MN)$. β is close to zero when $\hat{\mathbf{A}}_d^i$ approximates its true value $\mathbf{A}_d$. The detailed derivation can refer to Appendix B.

Define $\mathbf{D} = \sum_{q=0}^Q \mathbf{T} \text{diag}\{\mathbf{b}_q\} \mathbf{F}_{MN}^H \text{diag}\{\mathbf{F}_{MN \times L} \mathbf{c}_q\} \mathbf{F}_{MN} \mathbf{T}^H$. Note that $\mathbf{A}_d \mathbf{c}$ is equivalent to $\mathbf{D} \mathbf{x}_d$. $\hat{\mathbf{y}}_d^i$ is thus considered as a Gaussian random vector with mean of $\sqrt{1 - \alpha^*} \mathbf{D} \mathbf{x}_d$ and variance of $\sigma_i^2 \mathbf{I}_{MN}$ with $\sigma_i^2 = (\sigma_{e,i}^2 + \sigma_w^2 + \sigma_z^2)$, *i.e.*,

$$\hat{\mathbf{y}}_d^i = \sqrt{1 - \alpha^*} \mathbf{D} \mathbf{x}_d + \mathbf{e}^i + \mathbf{w} + \mathbf{z}. \quad (29)$$

By using BEM coefficient estimate, an estimate of $\mathbf{D}$ is

$$\hat{\mathbf{D}}^i = \sum_{q=0}^Q \mathbf{T} \text{diag}\{\mathbf{b}_q\} \mathbf{F}_{MN}^H \text{diag}\{\mathbf{F}_{MN \times L} \hat{\mathbf{c}}_q^i\} \mathbf{F}_{MN} \mathbf{T}^H. \quad (30)$$

By letting $\hat{\mathbf{G}}^i = \sqrt{1 - \alpha^*} \hat{\mathbf{D}}^i$, the *a posteriori* mean and variance of data $\mathbf{x}_d$ are given by

$$\begin{aligned}\mathbf{m}_{\text{pos}}^i &= \mathbf{V}_{\text{pos}}^i ((\hat{\mathbf{G}}^i)^H (\sigma_i^2)^{-1} \hat{\mathbf{y}}_d^i + \text{diag}\{\mathbf{v}_{\text{pri}}^{i-1}\}^{-1} \mathbf{m}_{\text{pri}}^{i-1}), \\ \mathbf{V}_{\text{pos}}^i &= ((\hat{\mathbf{G}}^i)^H (\sigma_i^2)^{-1} \hat{\mathbf{G}}^i + \text{diag}\{\mathbf{v}_{\text{pri}}^{i-1}\}^{-1})^{-1},\end{aligned}\quad (31)$$

where $\mathbf{v}_{\text{pri}}^{i-1}$ and $\mathbf{m}_{\text{pri}}^{i-1}$ are *a priori* mean and variance of data symbol, which are obtained by using *a posteriori* LLRs of channel decoder. The estimation of $\mathbf{v}_{\text{pri}}^{i-1}$ and $\mathbf{m}_{\text{pri}}^{i-1}$ can be obtained from (38). The extrinsic mean and variance of n -th ($n = 0, 1, \dots, MN - 1$) data symbol are thus acquired as

$$\begin{aligned}v_{\text{ext}}^i[n] &= ((\mathbf{V}_{\text{pos}}^i[n, n])^{-1} - (\mathbf{v}_{\text{pri}}^{i-1}[n])^{-1})^{-1}, \\ m_{\text{ext}}^i[n] &= v_{\text{ext},n}^i \left(\frac{\mathbf{m}_{\text{pos}}^i[n]}{\mathbf{V}_{\text{pos}}^i[n, n]} - \frac{\mathbf{m}_{\text{pri}}^{i-1}[n]}{\mathbf{v}_{\text{pri}}^{i-1}[n]} \right).\end{aligned}\quad (32)$$

3) *Demapping & Data Decoding*: This subsection aims to obtain the *a posteriori* and extrinsic LLRs of channel decoder after i -th iteration, denoted as $\mathbf{L}_{D,\text{pos}}^i$ and $\mathbf{L}_{D,\text{ext}}^i$, respectively. The two inputs of demapper are: a) extrinsic mean $m_{\text{ext}}^i[n]$ and variance $v_{\text{ext}}^i[n]$ of data symbol from signal detection; and b) *a priori* LLRs of coded bits $\mathbf{L}_{E,\text{pri}}^i$, which are obtained from the extrinsic LLRs of channel decoder after $(i-1)$ -th iteration denoted as $\mathbf{L}_{D,\text{ext}}^{i-1}$. *A priori* symbol probability of n -th data symbol $P_{\text{pri},\mathbf{x}_d[n]=\chi}^i$ is thus expressed as

$$P_{\text{pri},\mathbf{x}_d[n]=\chi}^i \propto \prod_{k=0}^{\log_2 K - 1} e^{-\chi[k] \mathbf{L}_{E,\text{pri}}^i[n \log_2 K + k]}, \quad (33)$$

where χ is the symbol from a certain modulation alphabet $\mathbb{K}$ whose modulation order is K and $\chi[k]$ indicates the k -th bit in χ . With $P_{\text{pri},\mathbf{x}_d[n]=\chi}^i$, the *a posteriori* symbol probability of n -th data symbol $P_{\text{pos},\mathbf{x}_d[n]=\chi}^i$ is then calculated by using

$$P_{\text{pos},\mathbf{x}_d[n]=\chi}^i \propto P_{\text{pri},\mathbf{x}_d[n]=\chi}^{i-1} \times e^{\frac{-|\chi - m_{\text{ext},n}^i|^2}{v_{\text{ext},n}^i}}. \quad (34)$$

With $P_{\text{pos},\mathbf{x}_d[n]=\chi}^i$, the *a posteriori* LLR of k -th bit of n -th data symbol is given by

$$\mathbf{L}_{E,\text{pos}}^i[n \log_2 K + k] = \log \frac{\sum_{\chi \in \mathbb{K}, \chi[k]=0} P_{\text{pos},\mathbf{x}_d[n]=\chi}^i}{\sum_{\chi \in \mathbb{K}, \chi[k]=1} P_{\text{pos},\mathbf{x}_d[n]=\chi}^i}. \quad (35)$$

The corresponding extrinsic LLR is expressed as

$$\mathbf{L}_{E,\text{ext}}^i = \mathbf{L}_{E,\text{pos}}^i - \mathbf{L}_{E,\text{pri}}^{i-1}, \quad (36)$$

which will be sent to deinterleaver and channel decoder. Note that in the initial stage $i = 0$, $\mathbf{L}_{E,\text{pri}}^0$ is set to $\mathbf{0}_{MN \log_2 K \times 1}$.

Since channel encoder and channel decoder are not the research focus of this paper, their technical details are not presented. Any existing channel encoders and decoders, *i.e.*, convolutional coding and LDPC coding [35], can be applied with proposed Turbo BEM OTFS scheme. After $\mathbf{L}_{E,\text{ext}}^i$ is passed through deinterleaver, channel decoder, and interleaver, the extrinsic and *a posteriori* LLRs are produced and denoted as $\mathbf{L}_{D,\text{ext}}^i$ and $\mathbf{L}_{D,\text{pos}}^i$, respectively. $\mathbf{L}_{D,\text{ext}}^i$ will be fed back to demapper as its *a priori* LLRs. As will be discussed next, $\mathbf{L}_{D,\text{pos}}^i$ will pass through mapper to calculate *a posteriori* mean and variance of data symbol, which will be utilized as *priori* information for channel estimation and signal detection to enhance their performance.

4) *Mapping*: This subsection aims to obtain the *a priori* mean and variance of n -th data symbol estimate after i -th iteration, denoted as $\mathbf{m}_{\text{pri}}^i[n]$ and $\mathbf{v}_{\text{pri}}^i[n]$, by using the *a posteriori* LLRs of channel decoder from i -th iteration $\mathbf{L}_{D,\text{pos}}^i$.

With $\mathbf{L}_{D,\text{pos}}^i$, the *a priori* probability of n -th data symbol is

$$P_{M,\text{pri},\mathbf{x}_d[n]=\chi}^i \propto \prod_{k=0}^{\log_2 K - 1} e^{-\chi[k] \mathbf{L}_{D,\text{pos}}^i[n \log_2 K + k]}. \quad (37)$$

Hence, *a priori* mean and variance of n -th data symbol estimate are expressed as

$$\begin{aligned}\mathbf{m}_{\text{pri}}^i[n] &= \sum_{\chi \in \mathbb{A}} P_{M,\text{pri},\mathbf{x}_d[n]=\chi}^i \chi, \\ \mathbf{v}_{\text{pri}}^i[n] &= \sum_{\chi \in \mathbb{A}} P_{M,\text{pri},\mathbf{x}_d[n]=\chi}^i |\chi|^2 - |\mathbf{m}_{\text{pri}}^i[n]|^2.\end{aligned}\quad (38)$$

Then, $\mathbf{m}_{\text{pri}}^i[n]$ is fed back to channel estimation, and both $\mathbf{m}_{\text{pri}}^i[n]$ and $\mathbf{v}_{\text{pri}}^i[n]$ are to signal detection, to further enhance system reliability. Note that in the initial stage $i = 0$, $P_{M,\text{pri},\mathbf{x}_d[n]=\chi}$ is assumed to have equal probability as $1/K$ for all candidates in modulation alphabet.

The aforementioned procedures from BEM channel estimation, pilot cancellation and signal detection, demapping, deinterleaving, channel decoding, interleaving, and mapping are repeated until either one of two following stopping criterion is satisfied: a) the difference of data estimate variance for two subsequent iterations is below the predefined threshold

Algorithm 2: Proposed Turbo BEM OTFS Receiver

-
- Input:** Received signal $\mathbf{y}$, superimposed pilots $\mathbf{x}_{\text{sp}}$
Output: *A Posteriori* LLRs of data bits $\mathbf{L}_{\text{D,pos}}^i$
- 1 **Initial BEM channel estimation:** Obtaining an initial BEM coefficient vector estimate $\hat{\mathbf{c}}^0$ using (24).
 - 2 **Initialization:** *A priori* mean and variance of n -th ($n = 0, 1, \dots, N-1$) OTFS data symbol are set to 0 and 1, i.e., $\mathbf{m}_{\text{pri}}^0[n] = 0$ and $\mathbf{v}_{\text{pri}}^0[n] = 1$. The *a priori* LLRs of demapper are set to zero, i.e., $\mathbf{L}_{\text{E,pri}}^0 = \mathbf{0}_{MN \log_2 K \times 1}$. Iteration index $i = 1$.
 - 3 **repeat**
 - 4 **Signal detection:** Obtaining extrinsic mean and variance of n -th ($n = 0, 1, \dots, N-1$) OTFS data symbol $m_{\text{ext}}^i[n]$ and $v_{\text{ext}}^i[n]$ by using (27)-(32).
 - 5 **Demapping:** Obtaining extrinsic LLRs of demapper $\mathbf{L}_{\text{E,ext}}^i$ using (33)-(36).
 - 6 **Data decoding:** Passing extrinsic LLRs of demapper to deinterleaver, channel decoder, and interleaver, and obtaining extrinsic LLRs and *a posteriori* LLRs of channel decoder, i.e., $\mathbf{L}_{\text{D,ext}}^i$ and $\mathbf{L}_{\text{D,pos}}^i$. The extrinsic LLRs of channel decoder are fed back to demapper as its priori LLRs, i.e., $\mathbf{L}_{\text{E,pri}}^i = \mathbf{L}_{\text{D,ext}}^i$.
 - 7 **Mapping:** Obtaining *a priori* mean and variance of n -th ($n = 0, 1, \dots, N-1$) OTFS data symbol $\mathbf{m}_{\text{pri}}^i[n]$ and $\mathbf{v}_{\text{pri}}^i[n]$ using (37) and (38). $\mathbf{m}_{\text{pri}}^i[n]$ and $\mathbf{v}_{\text{pri}}^i[n]$ are fed back to signal detection as its priori inputs. $\mathbf{m}_{\text{pri}}^i[n]$ is fed back to iterative BEM channel estimation as its pseudo pilots.
 - 8 **Iterative BEM channel estimation:** Obtaining an enhanced BEM coefficient vector estimate $\hat{\mathbf{c}}^i$ by using (25) and (26).
 - 9 $i = i + 1$.
 - 10 **until stopping criteria are satisfied;**
 - 11 **return:** $\mathbf{L}_{\text{D,pos}}^i$.
-

λ , i.e., $|\mathbf{v}_{\text{pri}}^i - \mathbf{v}_{\text{pri}}^{i-1}| < \lambda$; and b) the maximum number of iterations I is reached. Note that the main challenge of including decoding component in the proposed iterative receiver lies in the design of soft-in soft-out equalizer. Though Turbo equalization is well known in the literature, it is not straightforward to implement Turbo iterations between BEM channel estimation, signal detection, and data decoding in OTFS. The soft input and soft output information for channel estimation, signal detection, demapping, and channel decoding need to be properly designed. OTFS channel estimation and signal detection need to be specially developed to exploit soft input information and generate soft output information. To the best of the authors' knowledge, they are open research problems in OTFS literature. This subsection is looking at these problems and develops a novel Turbo BEM OTFS receiver. The proposed receiver is summarized in Algorithm 2.

IV. PERFORMANCE ANALYSIS

A. Spectral Efficiency Analysis

This subsection aims to investigate spectral efficiency of proposed Turbo BEM OTFS scheme. Define B as

channel bandwidth. Assuming bandwidth S is used for data transmission, the spectral efficiency is given by $\text{SE} = \frac{S}{B} \log_2(1 + \text{SINR})$. Because the proposed Turbo BEM OTFS scheme is based on superimposed pilots, payload data can be transmitted over the whole communication band B , i.e., $S = B$. Hence, the spectral efficiency of proposed Turbo BEM OTFS scheme is expressed as $\text{SE} = \log_2(1 + \text{SINR})$.

The SINR after initial BEM channel estimation by using superimposed pilots has been derived in Subsection III-A to optimize BEM order and superimposed pilot power ratio. Denote SINR_i ($i \geq 1$) as SINR of proposed Turbo BEM OTFS scheme after i -th iteration. Similarly to the derivation from (7) to (9), SINR_i is formulated as

$$\text{SINR}_i = \frac{\mathbb{E}\{(1-\alpha)(\hat{\mathbf{c}}^i)^H \mathbf{A}_d^H \mathbf{A}_d \hat{\mathbf{c}}^i\}}{\mathbb{E}\{(\tilde{\mathbf{c}}^i)^H \mathbf{A}^H \mathbf{A} \tilde{\mathbf{c}}^i + \mathbf{w}^H \mathbf{w} + \mathbf{z}^H \mathbf{z}\}}. \quad (39)$$

with $\tilde{\mathbf{c}}^i = \mathbf{c} - \hat{\mathbf{c}}^i$ being BEM coefficient estimation error from i -th ($i \geq 1$) iteration. According to [14], the MSE of BEM coefficient estimation error can be derived as: $\mathbb{E}\{\tilde{\mathbf{c}}^i(\tilde{\mathbf{c}}^i)^H\} = ((\hat{\mathbf{A}}^i)^H \hat{\mathbf{A}}^i)^{-1} \sigma_{\text{wz}}^2$. The numerator of (39) denoted as $\text{SINR}_{i,N}$ can be formulated as

$$\begin{aligned} \text{SINR}_{i,N} &= \mathbb{E}\{\text{Tr}\{(1-\alpha)(\mathbf{A}_d^H \mathbf{A}_d \hat{\mathbf{c}}^i(\hat{\mathbf{c}}^i)^H)\}\} \\ &= \mathbb{E}\{\text{Tr}\{(1-\alpha)(\mathbf{A}_d^H \mathbf{A}_d (\tilde{\mathbf{c}}^i(\tilde{\mathbf{c}}^i)^H + \mathbf{c}\mathbf{c}^H))\}\} \\ &= \text{Tr}\{(1-\alpha)\mathbf{A}_d^H \mathbf{A}_d ((\hat{\mathbf{A}}^i)^H \hat{\mathbf{A}}^i)^{-1} \sigma_{\text{wz}}^2\} + (1-\alpha)p. \end{aligned} \quad (40)$$

The denominator of (39) denoted as $\text{SINR}_{i,D}$ is expressed as

$$\begin{aligned} \text{SINR}_{i,D} &= \mathbb{E}\{\text{Tr}\{\mathbf{A}^H \mathbf{A} \tilde{\mathbf{c}}^i(\tilde{\mathbf{c}}^i)^H\}\} + MN\sigma_{\text{wz}}^2 \\ &= \text{Tr}\{\mathbf{A}^H \mathbf{A} ((\hat{\mathbf{A}}^i)^H \hat{\mathbf{A}}^i)^{-1} \sigma_{\text{wz}}^2\} + MN\sigma_{\text{wz}}^2. \end{aligned} \quad (41)$$

Hence, (39) is finally given by

$$\text{SINR}_i = \frac{\text{Tr}\{(1-\alpha)\mathbf{A}_d^H \mathbf{A}_d ((\hat{\mathbf{A}}^i)^H \hat{\mathbf{A}}^i)^{-1} \sigma_{\text{wz}}^2\} + (1-\alpha)p}{\text{Tr}\{\mathbf{A}^H \mathbf{A} ((\hat{\mathbf{A}}^i)^H \hat{\mathbf{A}}^i)^{-1} \sigma_{\text{wz}}^2\} + MN\sigma_{\text{wz}}^2}. \quad (42)$$

The spectral efficiency of proposed scheme after i -th iteration is $\text{SE}_i = \log_2(1 + \text{SINR}_i)$. Assuming perfect channel estimation with $\tilde{\mathbf{c}}^i \rightarrow \mathbf{0}$ and perfect BEM modeling with $\sigma_z \rightarrow 0$, (42) is upper bounded by $\text{SINR}_i \leq \text{SINR}_{\text{UB}}$, with

$$\text{SINR}_{\text{UB}} = \frac{\mathbb{E}\{(1-\alpha)\mathbf{c}^H \mathbf{A}_d^H \mathbf{A}_d \mathbf{c}\}}{\mathbb{E}\{\mathbf{w}^H \mathbf{w}\}} = \frac{(1-\alpha)p}{MN\sigma_{\text{w}}^2}. \quad (43)$$

Hence, the spectral efficiency of proposed scheme is bounded by $\text{SE}_{\text{UB}} = \log_2(1 + \text{SINR}_{\text{UB}})$. Only for sake of establishing the upper bound (43), perfect channel estimation has been used. This upper bound will be used as a benchmark in Figs. 8b and 13b to indicate the gap between the proposed scheme with practical channel estimation and genie-aided scheme (upper bound (43)). In addition, to the best of the authors' knowledge, the analysis of converged SINR in coded systems with iterative decoding is not possible and not tractable. The expression for SINR at i -th iteration SINR_i has been derived, which however is related to i -th estimate of data-related matrix $\hat{\mathbf{A}}^i$. $\hat{\mathbf{A}}^i$ is not analyzable and tractable in coded systems with iterative decoding. Therefore, similarly to existing SP-aided works, e.g., [2], initial SINR is used for optimization purpose in our work. Simulation results show that

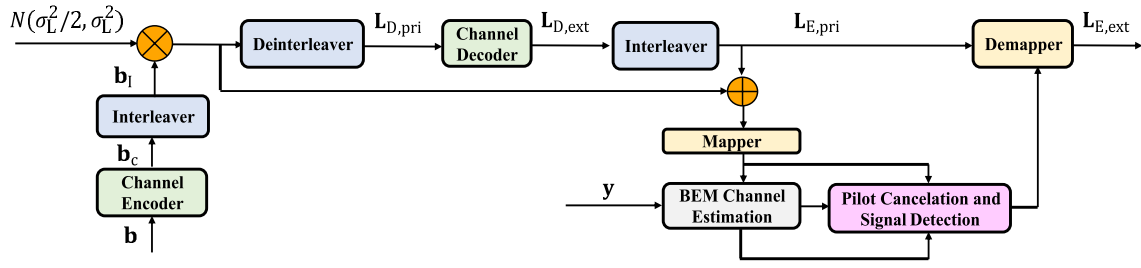


Fig. 6. Block diagram for generating EXIT chart of proposed Turbo BEM OTFS receiver.

the derived pilot power ratio using initial SINR yields the best mutual information and BERs in Figs. 11b and 11c.

B. EXIT Chart

EXIT chart is regarded as a helpful tool to investigate the transfer of mutual information and analyze the convergence behaviour of iterative receivers [23]. In this subsection, EXIT chart analysis of proposed Turbo BEM OTFS receiver is conducted to evaluate its convergence speed.

Fig. 6 shows the block diagram to generate the EXIT chart of proposed Turbo BEM OTFS receiver. Define $I_{D,in}$ as input mutual information of channel decoder. The *a priori* LLRs of channel decoder $L_{D,pri}\{b_c\}$ is typically modelled as an independent zero-mean Gaussian random vector with mean of $\sigma_L^2/2$ and variance of σ_L^2 [23], [36]. The value of σ_L is obtained by using $\sigma_L = J^{-1}(I_{D,in})$, where function $J(\cdot)$ is defined in (10) of [36]. The extrinsic LLRs of channel decoder is denoted as $L_{D,ext}\{b_c\}$. Its mutual information is calculated as $I_{D,out}$ by following the equations (12)-(15) in [36]. The *a priori* LLR of equalizer is obtained accordingly as $L_{E,pri}\{b_c\}$. By performing mapper, channel estimation, signal detection, and demapper, the extrinsic LLR of equalizer is calculated as $L_{E,ext}\{b_c\}$. With $L_{E,pri}\{b_c\}$ and $L_{E,ext}\{b_c\}$, the input and output mutual information of equalizer are obtained as $I_{E,in}$ and $I_{E,out}$ by using the equations in [36].

Following the simulation setting in Subsection V-A, the EXIT chart of proposed Turbo BEM OTFS receiver is plotted in Fig. 7 for $E_b/N_0 = 7.5$ dB and $E_b/N_0 = 10$ dB, respectively. As concluded in [23] and [36], the number of steps in the trajectory path of EXIT chart could be used to predict the convergence speed of iterative receivers. As can be observed from Fig. 7, at least 3 steps are required when the output mutual information of channel decoder $I_{D,out}$ changes from 0 to 1. As will be seen in Section V, Figs. 12a and 12b present the respective mean-square-error (MSE) of channel estimation and BER as a function of number of iterations. It can be seen that after 3 iterations the performance improvement either in terms of MSE of channel estimation and BER is negligible. Thus, the convergence speed of proposed Turbo BEM OTFS scheme through EXIT chart analysis in Fig. 7 matches with what will be observed from Fig. 12. Though Turbo receiver tends to incur higher latency due to iterative processing, EXIT chart analysis and simulation results show that the proposed receiver converges fast within 3 iterations. The latency of proposed receiver could decrease by reducing the number of iterations, at the potential expense of compromising reliability.

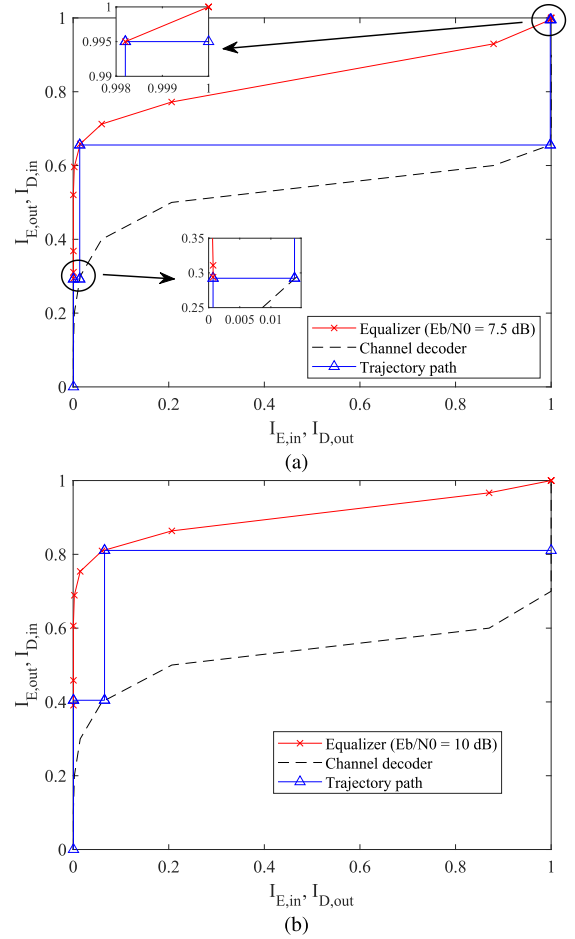


Fig. 7. EXIT chart with a trajectory path for proposed Turbo BEM OTFS receiver: a) $E_b/N_0 = 7.5$ dB and b) $E_b/N_0 = 10$ dB, with $M = 32$, $N = 16$, $f_c = 4$ GHz, $\Delta f = 15$ KHz, $L = 5$, $v = 250$ km/h, $\bar{f}_{max} = 0.06$, and Jakes Doppler spectrum.

C. Complexity Analysis

The computational complexity of proposed Turbo BEM OTFS receiver is investigated in this Subsection. They are summarized in three main aspects as shown in Table I, namely, BEM channel estimation, pilot cancellation and signal detection, and channel decoding. The complexities of initial and iterative BEM channel estimation are $\mathcal{O}(MNL^2Q^2 + MNLQ + L^3Q^3)$ and $\mathcal{O}((M^2N^2LQ + MNLQ + MNL^2Q^2 + L^3Q^3)I)$, respectively. Iterative BEM channel estimation dominates the overall complexity of BEM channel estimation. Pilot cancellation and signal detection has the complexity of

TABLE I

COMPUTATIONAL COMPLEXITY (M : NUMBER OF DELAY BINS, N : NUMBER OF DOPPLER BINS, L : CHANNEL LENGTH, Q : BEM ORDER, I : NUMBER OF ITERATIONS FOR PROPOSED TURBO BEM OTFS RECEIVER, AND I_D : NUMBER OF CHANNEL DECODING ITERATIONS)

Item		Computational complexity
BEM channel estimation	Initial stage	$\mathcal{O}(MNL^2Q^2 + MNLQ + L^3Q^3)$
	Iterative stage	$\mathcal{O}((M^2N^2LQ + MNLQ + MNL^2Q^2 + L^3Q^3)I)$
Pilot cancelation and signal detection	Full complexity	$\mathcal{O}((M^3N^3Q + M^2N^2Q + MNLQ)(I + 1))$
	Reduced complexity	$\mathcal{O}((M^2N^2Q + MNLQ)(I + 1))$
Channel decoding		$\mathcal{O}(M^2N^2I_D(I + 1))$ [35]

$\mathcal{O}((M^3N^3Q + M^2N^2Q + MNLQ)(I + 1))$. According to [35], the complexity of LDPC decoding per Turbo iteration is $\mathcal{O}(M^2N^2I_D)$, where I_D denotes the number of decoding iterations. Pilot cancelation and signal detection with a cubic complexity, i.e., $M^3N^3Q(I + 1)$, is the dominant item for the proposed Turbo BEM OTFS receiver. However, its complexity can be reduced by applying the approximation technique in Subsection V-B of [4] to (31). Specifically, the matrix inverse operation with a cubic complexity in (31) can be simplified by inverting the diagonal elements of approximated matrix, while preserving the similar performance. Because of BEM processing for Doppler-spread channel, the proposed receiver tends to have higher complexity than SP-aided OTFS scheme [2] in channel estimation and signal detection. Though SP-aided OTFS scheme [2] with delay-Doppler channel representation has lower complexity, its BER and spectral efficiency are worse than that of proposed receiver especially for Doppler-spread channel as will be seen in Figs. 8 and 13. The complexity of proposed receiver could decrease by reducing the number of iterations, at the expense of degraded reliability.

V. SIMULATION RESULTS

A. Simulation Setup

Monte Carlo simulations are carried out to evaluate the performance of proposed Turbo BEM OTFS scheme in comparison to subspace-aided OTFS scheme [1] and SP-aided OTFS scheme [2]. The simulation setting is set as follows, unless otherwise stated. The numbers of OTFS delay and Doppler bins are set to $M = 32$ and $N = 16$. The carrier frequency and subcarrier spacing are $f_c = 4$ GHz and $\Delta f = 15$ KHz. The 5G TDL-B channel with Jakes' Doppler spectrum and channel length $L = 5$ is adopted [26]. The detailed delay-power profile of 5G TDL-B channel can refer to Page 70 and 71 of [26]. The vehicle speed is 250 km/h for Figs. 8, 9a, 9b, 11, 12, and 14. The vehicle speed is 500 km/h for Figs. 9a, 9c, and 13. The modulation scheme is quadrature PSK (QPSK). LDPC coding with coding rate 1/2 is utilized. The number of Turbo iterations for proposed scheme is set to $I = 3$.

Tables II and III summarize the derived optimal superimposed pilot power ratio α^* and BEM order Q^* for various channels, including Doppler-shift channels with known/unknown Doppler shifts and Doppler-spread channels. It can be seen that α^* and Q^* change with channel, noise variance, velocity, etc. The impact of channel on the optimization of α^* and Q^* is reflected in BEM modeling error σ_z^2 and trace of channel correlation matrix p in (17). σ_z^2 and p may

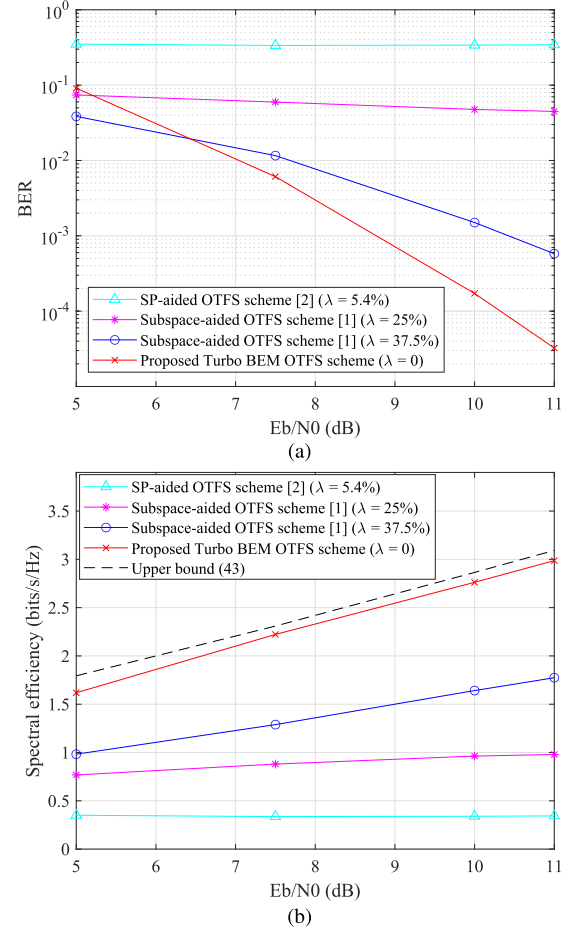


Fig. 8. a) BER, and b) spectral efficiency of proposed Turbo BEM OTFS, existing subspace-aided OTFS [1], and SP-aided OTFS [2] schemes, with $\bar{f}_{\max} = 0.06$, $v = 250$ km/h, and Jakes Doppler spectrum.

change with channels, and thus, α^* and Q^* are different at some cases.

B. Results and Discussion

Fig. 8 shows the BER and spectral efficiency of proposed Turbo BEM OTFS scheme, in comparison with the existing subspace-aided [1] and SP-aided [2] OTFS schemes, with $\bar{f}_{\max} = 0.06$ and $v = 250$ km/h. Subspace-aided [1] OTFS scheme is assisted by dedicated pilots. Unlike superimposed pilots, dedicated pilots imply that pilot symbols and data symbols are orthogonal in OTFS delay-Doppler domain. Fig. 1a of [2] and Fig. 9 of [1] are two examples of OTFS frame employing dedicated pilots. The dedicated pilot overhead is defined as $\lambda = N_p/(MN N_f)$, where N_p and N_f refer to the number

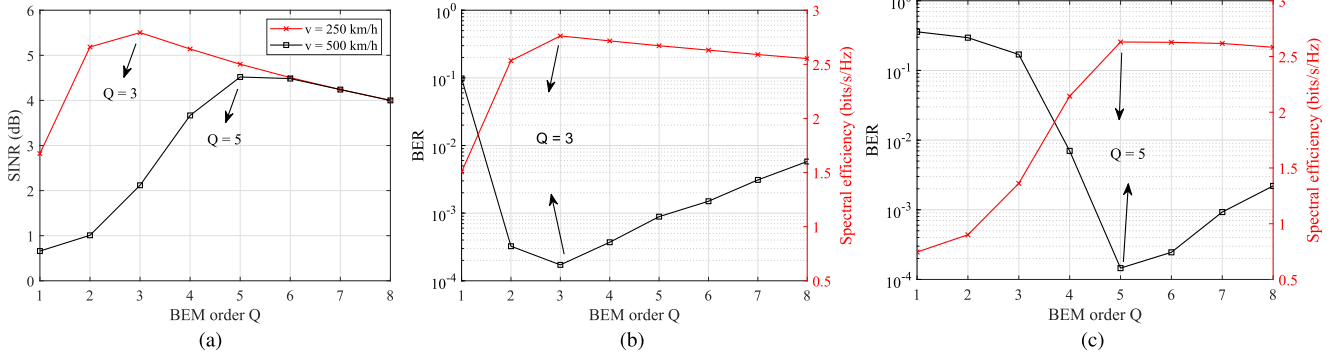


Fig. 9. a) Initial SINR as a function of BEM order Q for $v = 250$ km/h ($\bar{f}_{\max} = 0.06$) and $v = 500$ km/h ($\bar{f}_{\max} = 0.12$), b) BER and spectral efficiency as a function of BEM order Q for $v = 250$ km/h, and c) BER and spectral efficiency as a function of BEM order Q for $v = 500$ km/h, with Jakes Doppler spectrum, and $E_b/N_0 = 10$ dB. $Q^* = 3$ and $Q^* = 5$ are the optimal BEM order for $v = 250$ km/h and $v = 500$ km/h, respectively.

TABLE II

OPTIMAL SUPERIMPOSED PILOT POWER RATIO α^* AND BEM ORDER Q^* FOR DIFFERENT CHANNELS WITH $v = 250$ km/h ($\bar{f}_{\max} = 0.06$)

E_b/N_0 (dB)		5	7.5	10	11
Doppler-shift channel: known Doppler shifts	Q^*	0	0	0	0
	α^*	14.1%	19.1%	24.4%	26.8%
Doppler-shift channel: unknown Doppler shifts	Q^*	2	2	2	2
	α^*	20.4%	27.5%	34.6%	37.5%
Doppler-spread channel: Bell	Q^*	2	2	2	2
	α^*	20.2%	27.1%	33.9%	36.6%
Doppler-spread channel: Flat	Q^*	2	2	2	3
	α^*	19.7%	26.3%	32.5%	40.3%
Doppler-spread channel: Asymmetric Jakes	Q^*	2	3	3	3
	α^*	19.3%	29.7%	37.2%	40.2%
Doppler-spread channel: Jakes	Q^*	3	3	3	3
	α^*	22%	29.7%	37.2%	40.2%
Doppler-spread channel: Gaussian	Q^*	3	3	4	4
	α^*	20.9%	27.9%	38.5%	41.4%

TABLE III

OPTIMAL SUPERIMPOSED PILOT POWER RATIO α^* AND BEM ORDER Q^* FOR $v = 250$ km/h ($\bar{f}_{\max} = 0.06$) AND $v = 500$ km/h ($\bar{f}_{\max} = 0.12$), WITH JAKES DOPPLER SPECTRUM

E_b/N_0 (dB)		5	7.5	10	11
Doppler-spread channel: Jakes, $v = 250$ km/h	Q^*	3	3	3	3
	α^*	22%	29.7%	37.2%	40.2%
Doppler-spread channel: Jakes, $v = 500$ km/h	Q^*	5	5	5	5
	α^*	23.4%	31.7%	39.3%	42.1%

of orthogonal pilot symbols and number of OTFS frames. Two OTFS frames, one with dedicated pilots and another with superimposed pilots, are employed in [2]. Its dedicated pilots overhead is calculated as $\lambda = \frac{(2L+1)(4\lceil N\bar{f}_{\max} \rceil + 1)}{2MN} = 5.4\%$. Because the proposed scheme is assisted by one OTFS frame with superimposed pilots only, its dedicated pilot overhead is $\lambda = 0$. The subspace-aided OTFS scheme [1] uses one OTFS frame and its dedicated pilot overheads are set to $\lambda = 25\%$ and $\lambda = 37.5\%$. For a fair comparison, the (dedicated/superimposed) pilot power ratio of proposed Turbo BEM and existing [1], [2] OTFS systems are set to the same, whose values are shown in Table II.

As can be seen from Fig. 8a, the SP-aided [2] OTFS scheme has the worst BER, because it was designed for Doppler-shift channel. As discussed in Section II, delay-Doppler channel representation with a single Doppler shift

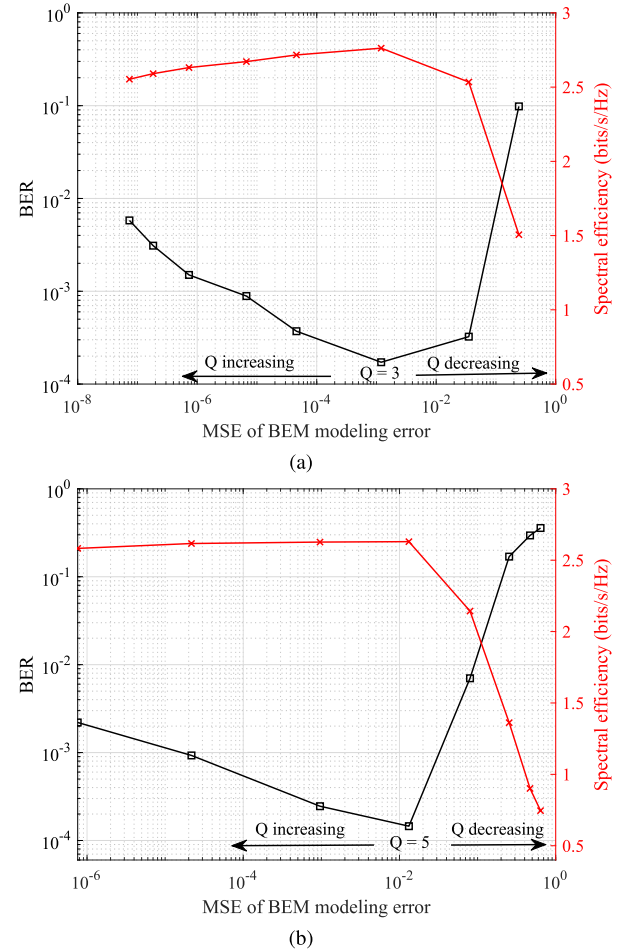


Fig. 10. Impact of BEM modeling error on BER and spectral efficiency of proposed scheme for a) $v = 250$ km/h ($\bar{f}_{\max} = 0.06$), and b) $v = 500$ km/h ($\bar{f}_{\max} = 0.12$), with Jakes Doppler spectrum, and $E_b/N_0 = 10$ dB.

for each path could not well characterize Doppler-spread channel with multiple Doppler shifts, more randomness and uncertainties. Thus, the method [2] based on it would not function well for Doppler-spread channel considered in this paper. The subspace-aided [1] OTFS scheme with $\lambda = 25\%$ has a BER error floor of 4×10^{-2} , and its BER enhances when λ increases to 37.5%. The proposed Turbo BEM OTFS

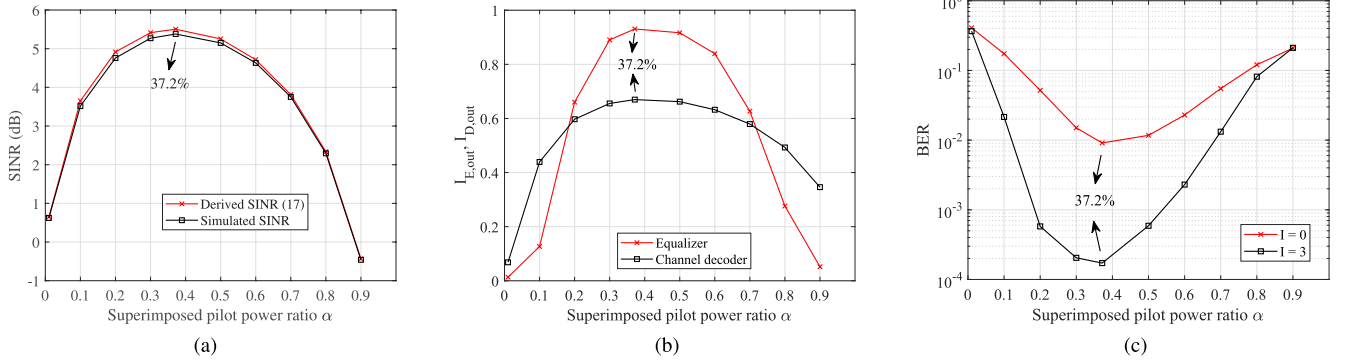


Fig. 11. a) Initial SINR, b) mutual information of equalizer $I_{E,out}$ and channel decoder $I_{D,out}$, and c) BER of proposed Turbo BEM OTFS scheme, as a function of superimposed pilot power ratio α , with $f_{\max} = 0.06$, $v = 250$ km/h, Jakes Doppler spectrum, and $E_b/N_0 = 10$ dB, where 37.2% is its optimal superimposed pilot power ratio.

scheme has superior BER over two existing schemes [1], [2], especially when E_b/N_0 is beyond 7.5 dB. The subspace-aided [1] scheme is assisted by dedicated pilots, and there is no interference between pilots and data. The proposed scheme is based on superimposed pilots, and the interference between pilots and data always exists. Due to data interference and big noise variance at low E_b/N_0 , channel estimation by using superimposed pilots cannot give a good initial channel estimate, and thus, the following data-aided channel estimation cannot further refine channel estimation. That is why the proposed scheme has inferior BER over the competitor [1] at low E_b/N_0 .

The spectral efficiency of proposed Turbo BEM OTFS scheme is calculated using (42) and shown in Fig. 8b, in comparison to the existing schemes [1], [2]. The proposed scheme has higher spectral efficiency over the existing schemes [1], [2] regardless of the value of E_b/N_0 . For example, at $E_b/N_0 = 11$ dB, the spectral efficiency of proposed scheme is approximately 1.7 times of that of existing scheme [1] with $\lambda = 37.5\%$. As mentioned previously, the SP-aided scheme [2] based on delay-Doppler channel representation with a single Doppler shift for each path does not work well for Doppler-spread channel. Hence, its spectral efficiency is worse than other schemes in Fig. 8b. The higher spectral efficiency of proposed scheme is because of the following two reasons. Assisted with superimposed pilots, the proposed scheme uses the whole bandwidth for data transmission, while the existing schemes [1], [2] reserves some exclusive bands for dedicated pilot transmission. On the other hand, the proposed scheme has superior BER (Fig. 8a) over the existing scheme [1], which thus leads to higher SINR and higher spectral efficiency. Last but not least, the spectral efficiency of proposed scheme approaches the upper bound (43) which is obtained by assuming perfect channel estimation.

Fig. 9 shows initial SINR, BER, and spectral efficiency of proposed scheme as a function of BEM order Q for $v = 250$ km/h and $v = 500$ km/h, with $E_b/N_0 = 10$ dB. According to Table III, the optimal BEM order is obtained as $Q^* = 3$ and $Q^* = 5$ for $v = 250$ km/h and $v = 500$ km/h, respectively, which yield the highest initial SINRs in Fig. 9a. Also, the derived optimal BEM order leads to the lowest BER and highest spectral efficiency in Figs. 9b and 9c. Fig. 10 shows the impact of BEM modeling error

on BER and spectral efficiency of proposed scheme. The modeling error should not be either too big or too small. The bigger BEM modeling error worsens BEM channel estimation, thus yielding poorer BER and spectral efficiency. The smaller BER modeling error requires a larger BEM order. It then incurs higher superimposed pilot power and leaves less power for data transmission, resulting in reduced BER and spectral efficiency as well. Since BEM modeling error is closely related to BEM order, the selection of a suitable BEM modeling error is equivalent to the selection of a suitable BEM order, which has been studied in Subsection III-A.

Fig. 11 shows initial SINR, mutual information, and BER of proposed Turbo BEM OTFS scheme as a function of superimposed pilot power ratio α , with $f_{\max} = 0.06$, $v = 250$ km/h, Jakes Doppler spectrum, and $E_b/N_0 = 10$ dB. The theoretical value of optimal α is calculated as $\alpha^* = 37.2\%$. It can be seen that $\alpha^* = 37.2\%$ yields the highest theoretical and simulated SINR in Fig. 11a, highest mutual information in Fig. 11b, and lowest BERs in Fig. 11c, which validates the effectiveness of proposed superimposed pilot power ratio optimization algorithm in Subsection III-A. Besides, it can be observed from Fig. 11a that the theoretical SINR derived in (17) is close to the simulated SINR, which verifies the accuracy of derivation in (17).

Fig. 12 studies the convergence speed of proposed Turbo BEM OTFS scheme in terms of MSE of channel estimation and BER. It can be seen that the proposed Turbo BEM OTFS scheme converges fast within 3 iterations. The performance enhancement after $I = 3$ is negligible. Thanks to the soft information exchange between channel estimation and channel decoder, the MSE of channel estimation and BER enhances greatly when the number of iterations is from $I = 0$ to $I = 3$. For example, the BER at $E_b/N_0 = 10$ dB decreases almost 100 times after 3 iterations. Moreover, the convergence speed results from Fig. 12 are consistent with that evaluated through EXIT chart analysis in Fig. 7 of Subsection IV-B.

Fig. 13 shows respectively the BER and spectral efficiency of proposed Turbo BEM OTFS scheme, existing subspace-aided [1], and SP-aided [2] OTFS schemes, with $f_{\max} = 0.12$, $v = 500$ km/h, and Jakes Doppler spectrum. The optimal superimposed pilot power ratio α^* is calculated and shown in Table III. Similarly to Fig. 8, the dedicated pilot overhead of proposed Turbo BEM OTFS schemes is $\lambda = 0$. The

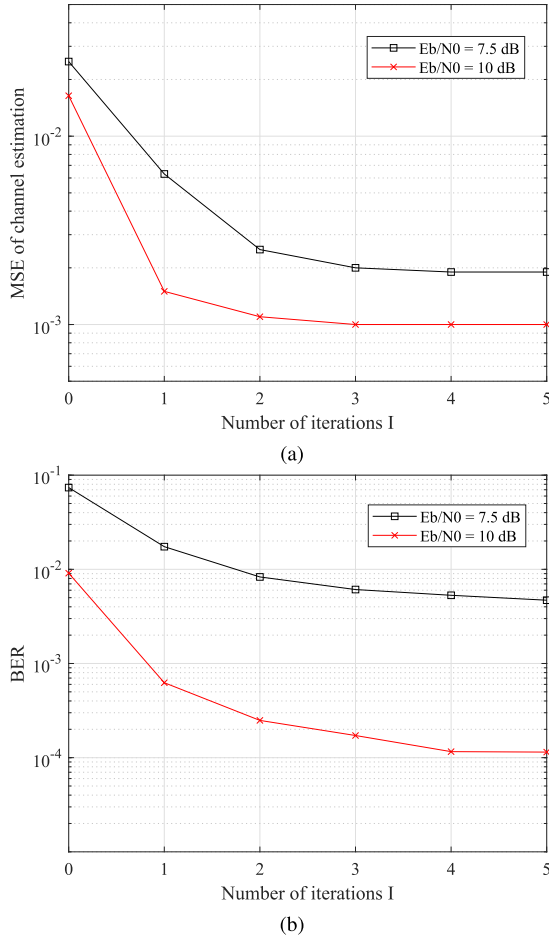


Fig. 12. a) MSE of channel estimation, and b) BER as a function of number of iterations I of proposed Turbo BEM OTFS scheme, with $\bar{f}_{\max} = 0.06$, $v = 250$ km/h, and Jakes Doppler spectrum.

pilot overhead of SP-aided OTFS scheme [2] is calculated as $\lambda = 9.7\%$. The subspace-aided OTFS scheme [1] has $\lambda = 25\%$ and $\lambda = 37.5\%$. For a fair comparison, the (dedicated/superimposed) pilot power ratio of proposed Turbo and existing [1], [2] OTFS schemes are set to the same, whose values are shown in Table III. The interference between data and superimposed pilots becomes more serious when velocity increases. While the existing subspace-aided OTFS scheme [1] is based on a large number of dedicated pilots and its estimation is free from data interference. That is why the proposed scheme is inferior than existing subspace-aided [1] OTFS scheme in terms of BER when E_b/N_0 is below 7.5 dB. Nevertheless, the proposed OTFS scheme has better BER when E_b/N_0 is above 7.5 dB, and an E_b/N_0 gain of 2 dB can be achieved. Besides, as can be observed from Fig. 13b, the proposed scheme has higher spectral efficiency than existing [1], [2] schemes. Likewise in Fig. 8, the SP-aided scheme [2] underperforms other schemes, because it is limited to Doppler-shift channel.

Fig. 14 shows BER of proposed Turbo BEM OTFS scheme for various channels, with $\bar{f}_{\max} = 0.06$ and $v = 250$ km/h. Their corresponding optimal superimposed pilot power ratios α^* are calculated and shown in Table II. Due to Doppler diversity provided by Doppler-spread channels, they exhibit lower BER than Doppler-shift channels. Moreover, it can

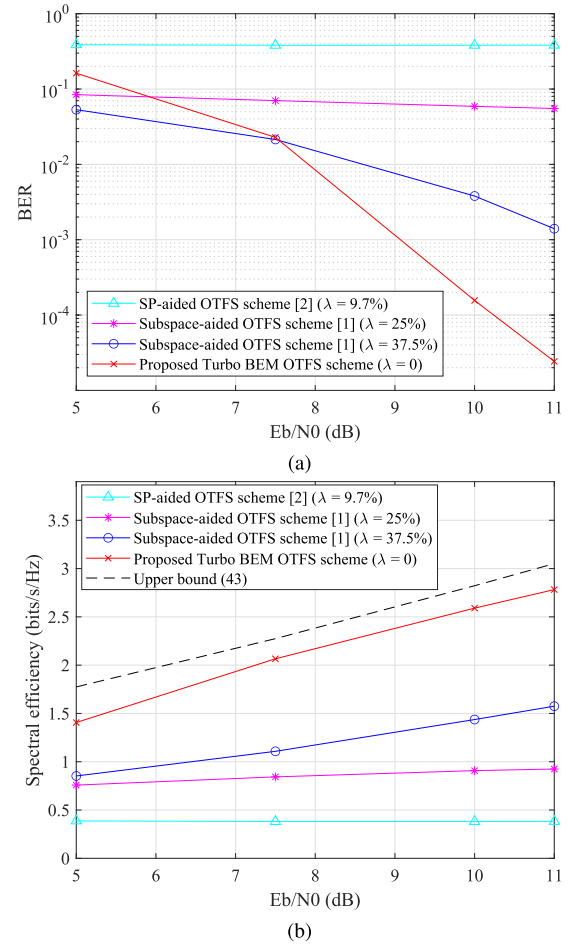


Fig. 13. a) BER, and b) spectral efficiency of proposed Turbo BEM OTFS scheme and existing subspace-aided [1] and SP-aided [2] OTFS schemes, with $\bar{f}_{\max} = 0.12$ and $v = 500$ km/h, and Jakes Doppler spectrum.

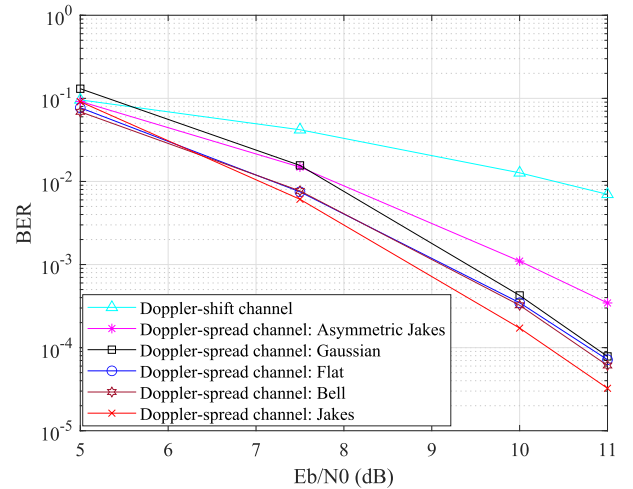


Fig. 14. BER of proposed Turbo BEM OTFS scheme for various types of channels, with $\bar{f}_{\max} = 0.06$ and $v = 250$ km/h.

be seen that Jakes spectrum has the best BER. This is because Doppler-spread channels with Jakes spectrum has a wide Doppler spread with peak power at maximum Doppler frequency, and thus, achieves higher Doppler diversity and lower BER. The inferior BER of asymmetric Jakes spectrum is because of its halved Doppler spread compared to other Doppler-spread channels. The Gaussian-shaped, flat-shaped,

and bell-shaped Doppler spectrum are found to have the similar BER.

VI. CONCLUSION

In this paper, a Turbo BEM OTFS scheme, with superimposed pilot power ratio and BEM order optimization and Turbo BEM OTFS receiver, has been proposed for high-mobility communications subject to either Doppler-shift or Doppler-spread channel. By using BEM to represent channels, the proposed scheme is applicable to various channels (*e.g.*, Doppler-shift channels, Doppler-spread channels with various types of Doppler spectrum), while SP-aided OTFS scheme [2] applies to Doppler-shift channel only. The BEM order is specially optimized for each channel, instead of simply increasing or reducing it. The derived SINR is close to the simulated SINR, and the derived optimal superimposed pilot power ratio and BEM order lead to the highest SINR, highest mutual information, and lowest BER. By utilizing superimposed pilots, the proposed scheme has higher spectral efficiency than subspace-aided [1] and SP-aided [2] OTFS schemes. By exchanging soft information from channel estimation, signal detection, to data detection, it has superior BER over the existing schemes [1], [2], and can save E_b/N_0 of 2 dB. EXIT chart analysis and simulation results also validate its fast convergence speed of 3 iterations.

APPENDIX A

DERIVATION OF $\text{Tr}\{\mathbb{E}\{\mathbf{A}_d^H \mathbf{A}_d \mathbf{c} \mathbf{c}^H\}\}$

Define $\mathbf{h}$ as time-varying channel impulse response vector of length $MN(L+1)$. $\mathbf{h}$ is given by $\mathbf{h} = [\mathbf{h}_0[0], \mathbf{h}_1[0], \dots, \mathbf{h}_L[0], \mathbf{h}_0[1], \mathbf{h}_1[1], \dots, \mathbf{h}_L[1], \dots, \mathbf{h}_0[MN-1], \mathbf{h}_1[MN-1], \dots, \mathbf{h}_L[MN-1]]$. The true BEM coefficient

$\mathbf{c}$ is thus given by $\mathbf{c} = \mathbf{E}\mathbf{h}$, with $\mathbf{E} = (\mathbf{B} \otimes \mathbf{I}_{L+1})^\dagger$. Due to the property of the inverse of a Kronecker product, $\mathbf{E}$ is given by $\mathbf{E} = \mathbf{B}^\dagger \otimes \mathbf{I}_{L+1}$. Define $\mathbf{A}_{d,q,l'} = (\mathbf{F}_N \otimes \mathbf{I}_M) \text{diag}\{\mathbf{b}_q\} \mathbf{f}_l'$, with $\mathbf{f}_l' = \mathbf{F}_{MN}^H \text{diag}\{\mathbf{F}_{MN}(\mathbf{F}_N^H \otimes \mathbf{I}_M) \mathbf{x}_d\} \mathbf{F}_{MN}(:, l')$. Due to the property of diagonal operator, $\mathbf{A}_{d,q,l'}$ can be written as $\mathbf{A}_{d,q,l'} = (\mathbf{F}_N \otimes \mathbf{I}_M) \text{diag}\{\mathbf{f}_l'\} \mathbf{b}_q$. Define $\mathbf{A}_{d,l'} = [\mathbf{A}_{d,0,l'}, \mathbf{A}_{d,1,l'}, \dots, \mathbf{A}_{d,Q,l'}]$, which is expressed as $\mathbf{A}_{d,l'} = (\mathbf{F}_N \otimes \mathbf{I}_M) \text{diag}\{\mathbf{f}_l'\} \mathbf{B}$. As a result, $\mathbf{A}_d$ can be given by $\mathbf{A}_d = (\mathbf{F}_N \otimes \mathbf{I}_M) \mathbf{G}(\mathbf{B} \otimes \mathbf{I}_{L+1})$, where $\mathbf{G} = [\text{diag}\{\mathbf{f}_0\}, \text{diag}\{\mathbf{f}_1\}, \dots, \text{diag}\{\mathbf{f}_L\}]$. $\mathbf{G}^H \mathbf{G}$ can be obtained as (44), shown at the bottom of the page.

Because of the property of DFT matrix $\mathbf{F}_{MN}$, $\mathbf{G}^H \mathbf{G}$ is expressed as $\mathbf{G}^H \mathbf{G} \approx \mathbf{I}_{MN} \otimes \mathbf{I}_{L+1}$. Hence, $\text{Tr}\{\mathbb{E}\{\mathbf{A}_d^H \mathbf{A}_d \mathbf{c} \mathbf{c}^H\}\}$ in (15) is further expressed as (45), shown at the bottom of the page. Because $\mathbf{F}_N^H \mathbf{F}_N = \mathbf{I}_N$, $\mathbf{I}_N \otimes \mathbf{I}_M = \mathbf{I}_{MN}$, and $\mathbf{G}^H \mathbf{G} \approx \mathbf{I}_{MN} \otimes \mathbf{I}_{L+1}$, (45) is rewritten as (46), shown at the bottom of the page. Because $\mathbf{B}^\dagger \mathbf{B} = \mathbf{I}_{(L+1)Q}$ and the property of trace, (46) is finally given by

$$\text{Tr}\{\mathbb{E}\{\mathbf{A}_d^H \mathbf{A}_d \mathbf{c} \mathbf{c}^H\}\} = \text{Tr}\{\mathbf{R}_{hh}\}. \quad (47)$$

APPENDIX B

DISTRIBUTION OF $\mathbf{e}^i$

The BEM coefficient vector is given by $\mathbf{c} = (\mathbf{B} \otimes \mathbf{I}_{L+1})^\dagger \mathbf{h}$. Note that channel is Rayleigh fading and $\mathbf{h}$ is a complex Gaussian random vector with zero mean [25], [26]. Because $(\mathbf{B} \otimes \mathbf{I}_{L+1})^\dagger$ is deterministic, the BEM coefficient vector $\mathbf{c}$ with a multiplication of a deterministic matrix and a Gaussian random vector can also be considered as a Gaussian random

$$\mathbf{G}^H \mathbf{G} = \begin{bmatrix} \text{diag}\{\mathbf{f}_0^H\} \text{diag}\{\mathbf{f}_0\} & \text{diag}\{\mathbf{f}_0^H\} \text{diag}\{\mathbf{f}_1\} & \dots & \text{diag}\{\mathbf{f}_0^H\} \text{diag}\{\mathbf{f}_L\} \\ \text{diag}\{\mathbf{f}_1^H\} \text{diag}\{\mathbf{f}_0\} & \text{diag}\{\mathbf{f}_1^H\} \text{diag}\{\mathbf{f}_1\} & \dots & \text{diag}\{\mathbf{f}_1^H\} \text{diag}\{\mathbf{f}_L\} \\ \vdots & \vdots & \ddots & \vdots \\ \text{diag}\{\mathbf{f}_L^H\} \text{diag}\{\mathbf{f}_0\} & \text{diag}\{\mathbf{f}_L^H\} \text{diag}\{\mathbf{f}_1\} & \dots & \text{diag}\{\mathbf{f}_L^H\} \text{diag}\{\mathbf{f}_L\} \end{bmatrix}. \quad (44)$$

$$\begin{aligned} \text{Tr}\{\mathbb{E}\{\mathbf{A}_d^H \mathbf{A}_d \mathbf{c} \mathbf{c}^H\}\} &= \text{Tr}\{\mathbb{E}\{\mathbf{A}_d^H \mathbf{A}_d \mathbf{D} \mathbf{h} \mathbf{h}^H \mathbf{D}^H\}\} \\ &= \text{Tr}\{\mathbb{E}\{((\mathbf{F}_N \otimes \mathbf{I}_M) \mathbf{G}(\mathbf{B} \otimes \mathbf{I}_{L+1}))^H (\mathbf{F}_N \otimes \mathbf{I}_M) \mathbf{G}(\mathbf{B} \otimes \mathbf{I}_{L+1}) (\mathbf{B}^\dagger \otimes \mathbf{I}_{L+1}) \mathbf{h} \mathbf{h}^H (\mathbf{B}^\dagger \otimes \mathbf{I}_{L+1})^H\}\} \\ &= \text{Tr}\{(\mathbf{B}^H \otimes \mathbf{I}_{L+1}) \mathbf{G}^H (\mathbf{F}_N^H \otimes \mathbf{I}_M) (\mathbf{F}_N \otimes \mathbf{I}_M) \mathbf{G}(\mathbf{B} \otimes \mathbf{I}_{L+1}) (\mathbf{B}^\dagger \otimes \mathbf{I}_{L+1}) \mathbb{E}\{\mathbf{h} \mathbf{h}^H\} ((\mathbf{B}^\dagger \otimes \mathbf{I}_{L+1}))^H\} \\ &= \text{Tr}\{(\mathbf{B}^H \otimes \mathbf{I}_{L+1}) \mathbf{G}^H ((\mathbf{F}_N^H \mathbf{F}_N) \otimes \mathbf{I}_M) \mathbf{G}(\mathbf{B} \otimes \mathbf{I}_{L+1}) (\mathbf{B}^\dagger \otimes \mathbf{I}_{L+1}) \mathbf{R}_{hh} (\mathbf{B}^\dagger \otimes \mathbf{I}_{L+1})^H\}. \end{aligned} \quad (45)$$

$$\begin{aligned} \text{Tr}\{\mathbb{E}\{\mathbf{A}_d^H \mathbf{A}_d \mathbf{c} \mathbf{c}^H\}\} &= \text{Tr}\{(\mathbf{B}^H \otimes \mathbf{I}_{L+1}) (\mathbf{B} \otimes \mathbf{I}_{L+1}) (\mathbf{B}^\dagger \otimes \mathbf{I}_{L+1}) \mathbf{R}_{hh} (\mathbf{B}^\dagger \otimes \mathbf{I}_{L+1})^H\} \\ &= \text{Tr}\{(\mathbf{B} \otimes \mathbf{I}_{L+1}) (\mathbf{B}^\dagger \otimes \mathbf{I}_{L+1}) \mathbf{R}_{hh} ((\mathbf{B}^\dagger \otimes \mathbf{I}_{L+1}))^H (\mathbf{B}^H \otimes \mathbf{I}_{L+1})\} \\ &= \text{Tr}\{(\mathbf{B} \mathbf{B}^\dagger) \otimes \mathbf{I}_{L+1} \mathbf{R}_{hh} ((\mathbf{B}^\dagger)^H \mathbf{B}^H) \otimes \mathbf{I}_{L+1}\}. \end{aligned} \quad (46)$$

$$\tilde{\mathbf{c}}^i (\tilde{\mathbf{c}}^i)^H = \begin{cases} (\mathbf{A}_{sp}^H \mathbf{A}_{sp})^{-1} \frac{\sigma_w^2 + \sigma_z^2 + 1 - \alpha}{\alpha} & i = 0 \\ ((\sqrt{\alpha^*} \mathbf{A}_{sp} + \sqrt{1 - \alpha^*} \hat{\mathbf{A}}_d^i)^H (\sqrt{\alpha^*} \mathbf{A}_{sp} + \sqrt{1 - \alpha^*} \hat{\mathbf{A}}_d^i))^{-1} (\sigma_w^2 + \sigma_z^2) & i \geq 1 \end{cases}. \quad (49)$$

$$\sigma_{e,i}^2 = \text{Tr}\{\alpha^* \mathbf{A}_{sp} ((\sqrt{\alpha^*} \mathbf{A}_{sp} + \sqrt{1 - \alpha^*} \hat{\mathbf{A}}_d^i)^H (\sqrt{\alpha^*} \mathbf{A}_{sp} + \sqrt{1 - \alpha^*} \hat{\mathbf{A}}_d^i))^{-1} (\sigma_w^2 + \sigma_z^2) \mathbf{A}_{sp}^H\} / (MN). \quad (51)$$

vector. In addition, the BEM coefficient estimate is given by $\hat{\mathbf{c}}^i = (\hat{\mathbf{A}}^{i-1})^\dagger \mathbf{y} = (\hat{\mathbf{A}}^{i-1})^\dagger (\mathbf{A}\mathbf{c} + \mathbf{w} + \mathbf{z})$. $\hat{\mathbf{A}}^{i-1}$ and $\mathbf{A}$ are deterministic, while the BEM coefficient $\mathbf{c}$, noise $\mathbf{w}$, and modeling error $\mathbf{z}$ are Gaussian random vectors with zero mean. The detailed description on distribution of modeling error can be found in [14]. Hence, $\hat{\mathbf{c}}^i$ is also a Gaussian random vector with zero mean. $(\mathbf{c} - \hat{\mathbf{c}}^i)$ is also Gaussian because the linear combination of two Gaussian random variables (vectors) is Gaussian. Because $\mathbf{A}_{\text{sp}}$ is deterministic, $\mathbf{e}^i = \sqrt{\alpha^*} \mathbf{A}_{\text{sp}} (\mathbf{c} - \hat{\mathbf{c}}^i)$ with a multiplication of a deterministic matrix and a Gaussian random vector is also a Gaussian random vector. The variance of $\mathbf{e}^i$ is denoted as $\sigma_{\mathbf{e},i}^2$ and derived as shown below

$$\begin{aligned} \sigma_{\mathbf{e},i}^2 &= \frac{\text{Tr}\{\alpha^* \mathbf{A}_{\text{sp}} (\mathbf{c} - \hat{\mathbf{c}}^i) (\mathbf{c} - \hat{\mathbf{c}}^i)^H \mathbf{A}_{\text{sp}}^H\}}{MN} \\ &= \frac{\text{Tr}\{\alpha^* \mathbf{A}_{\text{sp}}^H \mathbf{A}_{\text{sp}} (\mathbf{c} - \hat{\mathbf{c}}^i) (\mathbf{c} - \hat{\mathbf{c}}^i)^H\}}{MN}. \end{aligned} \quad (48)$$

According to [14], $\tilde{\mathbf{c}}^i (\tilde{\mathbf{c}}^i)^H$ is given by (49), shown at the bottom of the previous page.

For $i = 0$, (48) is further given by

$$\begin{aligned} \sigma_{\mathbf{e},0}^2 &= \frac{\text{Tr}\{\alpha^* \mathbf{A}_{\text{sp}}^H \mathbf{A}_{\text{sp}} (\mathbf{A}_{\text{sp}}^H \mathbf{A}_{\text{sp}})^{-1} \frac{\sigma_w^2 + \sigma_z^2 + 1 - \alpha^*}{\alpha^*}\}}{MN} \\ &= \frac{(L+1)(Q+1)(\sigma_w^2 + \sigma_z^2 + 1 - \alpha^*)}{MN}. \end{aligned} \quad (50)$$

For $i \geq 1$, (48) is expressed as (51), shown at the bottom of the previous page. The right-hand side of (51) is set to β . As a result, $\sigma_{\mathbf{e},i}^2$ is given by

$$\sigma_{\mathbf{e},i}^2 = \begin{cases} \frac{(L+1)(Q+1)(\sigma_w^2 + \sigma_z^2 + 1 - \alpha^*)}{MN} & i = 0 \\ \beta & i \geq 1 \end{cases}. \quad (52)$$

Note that β is close to zero, when $\hat{\mathbf{A}}_d^i$ approximates its true value $\mathbf{A}_d$.

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